

Further studies on Geometric Conservation Law and applications to high-order finite difference schemes with stationary grids

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ABSTRACT

The metrics and Jacobian in the fluid motion governing equations under curvilinear coordinate system have a variety of equivalent differential forms, which may have different discretization errors with the same difference scheme. The discretization errors of metrics and Jacobian may cause serious computational instability and inaccuracy in numerical results, especially for high-order finite difference schemes. It has been demonstrated by many researchers that the Geometric Conservation Law (GCL) is very important for high-order Finite Difference Methods (FDMs), and a proper form of metrics and Jacobian, which can satisfy the GCL, can considerably reduce discretization errors and computational instability. In order to satisfy the GCL for FDM, we have previously developed a Conservative Metric Method (CMM) to calculate the metrics [1] and the difference scheme δ^3 in the CMM is determined with the suggestion $\delta^3 = \delta^2$. In this paper, a Symmetrical Conservative Metric Method (SCMM) is newly proposed based on the discussions of the metrics and Jacobian in FDM from geometry viewpoint by following the concept of vectorized surface and cell volume in Finite Volume Methods (FVMs). Interestingly, the expressions of metrics and Jacobian obtained by using the SCMM with second-order central finite difference scheme are equivalent to the vectorized surfaces and cell volumes, respectively. The main advantage of SCMM is that it makes the calculations based on high-order WCNS schemes around complex geometry flows possible and somewhat easy. Numerical tests on linear and nonlinear problems indicate that the quality of numerical results may be largely enhanced by utilizing the SCMM, and the advantage of the SCMM over other forms of metrics and Jacobian may be more evident on highly nonuniform grids.

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1. Introduction

In our previous paper, a Conservative Metric Method (CMM) has been proposed in order to ensure the Geometry Conservation Law (GCL) for high-order finite difference schemes [1]. The CMM is achieved by computing grid metric derivatives through a conservative form with the same scheme applied for fluxes. It has also been shown that the CMM is suitable

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not only to interior schemes but also to boundary and near boundary schemes on the condition that the difference operators δ^1 for the first-order derivatives of fluxes are linear ones. A special case of the CMM can be found in Visbal and Gaitonde's numerical technique [12], where the same central compact scheme is applied to both flow fluxes and metrics. Visbal and Gaitonde's numerical technique was also followed by Nonomura et al. [8] in implementing the Weighted Compact Nonlinear Scheme (WCNS) derived by Deng and Zhang [15] on curvilinear grids. Nonomura et al. [8] also showed that the free-stream and vortex preservation properties of WCNS is superior to that of the WENO scheme [9], and the essential reason analysed by Deng et al. [1] is that CMM can be introduced to satisfy the GCL with WCNS. Although the inner-level difference operators δ^3 in the conservative metrics have no effect on the GCL, the constraint $\delta^3 = \delta^2$ is recommended by Deng et al. [1] where δ^2 is the outer-level difference operators in the conservative metrics. Deng et al. [1] also showed that extra errors in metrics may appear if $\delta^3 \neq \delta^2$.

In this paper, we further study the GCL from the geometry viewpoint. By analyzing the surface calculation of Finite Volume Method (FVM), we obtained a general metrics calculation method for Finite Difference Method (FDM), i.e., the metrics calculation of FDM for second order accuracy is equivalent to the surface calculation of FVM. This method is extended to high-order metrics calculation which not only satisfies CMM, but also requires $\delta^3 = \delta^2$. Thus we know why the errors may be large as $\delta^3 \neq \delta^2$ in [1]. Beyond the metrics, the calculation of Jacobian introduced by the transformation from Cartesian coordinates to curvilinear coordinates is traditionally only considered for moving or deforming grids as the Volume Conservation Law (VCL) [5,6], very little attention is paid for time-constant grids. In this paper, it will be verified that different expressions of the Jacobian may have quite different influences on numerical results with irregular grids. With an eye to the Jacobian obtained by a second-order central different scheme, from geometry viewpoint, we also expect that the Jacobian might describe the cell volume enclosed by its vectorized surfaces and should be uniquely determined by its vectorized surfaces as that in FVM. Unfortunately, the traditional definition of the Jacobian cannot be enclosed by its corresponding vectorized surfaces, though many endeavors are done to develop appropriate metrics. In the present work, two new symmetrical forms of Jacobian are derived from the geometry viewpoint, one is non-conservative, the other is conservative provided that the Surface Conservative Law (SCL) has already been satisfied. Special attentions should be paid to the symmetrical conservative Jacobian, because it can produce a volume-like Jacobian which is completely enclosed by its vectorized surfaces when the second-order central different scheme is used. Furthermore, the errors introduced by this conservative form are smaller than that by other forms as we will see through the numerical tests with irregular grids. Based on the analysis, a new Symmetrical Conservative Metric Method (SCMM) is derived. It is proved that the SCMM is equivalent to the vectorized surfaces and volume in second order accuracy as that in Finite Volume Method (FVM) in second order accuracy, thus the calculations with SCMM by FDM may be as robust as that by FVM. Furthermore, the main advantage of SCMM is that it makes the high-order WCNS schemes [15,18] calculations around complex geometry flows possible and somewhat easy.

In Section 2, the discretizations of governing equations by FDM and FVM with second-order accuracy is presented, and the SCL in both FDM and FVM are briefly introduced. The surface and volume in FDM and FVM are discussed from geometry viewpoint in Section 3, and SCMM is proposed via the discussions. In Section 4 the implementation of the SCMM on high-order procedure is presented in details. Some numerical tests including the linear wave propagation problems and nonlinear isentropic vortex moving problems are simulated with different forms of metrics and Jacobian on randomized grids, and the numerical results are discussed in Section 5 with a test on a typical flow around ONERA M6 wing. Finally, concluding remarks are presented in Section 6.

2. Discretizations of governing equations in FDM and FVM

2.1. Discretizations of governing equations in FDM and SCL

The strong conservative three-dimensional Euler equation in Cartesian coordinates (t, x, y, z) are

$$\frac{\partial Q}{\partial t} + \frac{\partial E}{\partial x} + \frac{\partial F}{\partial y} + \frac{\partial G}{\partial z} = 0, \quad (1)$$

where Q is dependent variable and E, F and G are all convective fluxes

$$Q = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ \rho e \end{pmatrix}, \quad E = \begin{pmatrix} \rho u \\ \rho uu + p \\ \rho vu \\ \rho wu \\ (\rho e + p)u \end{pmatrix}, \quad F = \begin{pmatrix} \rho v \\ \rho uv \\ \rho vv + p \\ \rho wv \\ (\rho e + p)v \end{pmatrix}, \quad G = \begin{pmatrix} \rho w \\ \rho uw \\ \rho vw \\ \rho ww + p \\ (\rho e + p)w \end{pmatrix},$$

where

$$p = \frac{1}{\gamma M_\infty^2} \rho T, \quad e = \frac{p}{(\gamma - 1)\rho} + \frac{u^2 + v^2 + w^2}{2}$$

and γ is the specific heat ratio. The non-dimensional variables are defined as $\rho = \rho^*/\rho_\infty^*$, $(u, v, w) = (u^*, v^*, w^*)/U_\infty^*$, $T = T^*/T_\infty^*$ and $p = p^*/(\rho_\infty^* U_\infty^{*2})$, and $M_\infty = U_\infty^*/\sqrt{\gamma RT_\infty^*}$ is the free-stream Mach number.

Transformation from Cartesian coordinates (t, x, y, z) to curvilinear coordinates (τ, ξ, η, ζ) is

$$\begin{cases} \tau = t, \\ \xi = \xi(t, x, y, z), \\ \eta = \eta(t, x, y, z), \\ \zeta = \zeta(t, x, y, z) \end{cases} \quad (2)$$

and Eq. (1) in three-dimensional curvilinear coordinates (τ, ξ, η, ζ) are

$$\frac{\partial \hat{Q}}{\partial \tau} + \frac{\partial \hat{E}}{\partial \xi} + \frac{\partial \hat{F}}{\partial \eta} + \frac{\partial \hat{G}}{\partial \zeta} = 0, \quad (3)$$

where

$$\begin{aligned} \hat{Q} &= J^{-1} Q, \\ \hat{E} &= \hat{\xi}_t Q + \hat{\xi}_x E + \hat{\xi}_y F + \hat{\xi}_z G, \\ \hat{F} &= \hat{\eta}_t Q + \hat{\eta}_x E + \hat{\eta}_y F + \hat{\eta}_z G, \\ \hat{G} &= \hat{\zeta}_t Q + \hat{\zeta}_x E + \hat{\zeta}_y F + \hat{\zeta}_z G. \end{aligned}$$

Here, the traditional forms of Jacobian and metrics are all listed as follows.

$$J^{-1} = \left| \frac{\partial(x, y, z)}{\partial(\xi, \eta, \zeta)} \right| = x_\xi y_\eta z_\zeta - x_\zeta y_\xi z_\eta + x_\eta y_\zeta z_\xi - x_\xi y_\zeta z_\eta + x_\zeta y_\xi z_\eta - x_\xi y_\eta z_\zeta, \quad (4)$$

$$\begin{aligned} \hat{\xi}_x &= J^{-1} \xi_x = y_\eta z_\zeta - y_\zeta z_\eta, & \hat{\xi}_y &= J^{-1} \xi_y = z_\eta x_\zeta - z_\zeta x_\eta, & \hat{\xi}_z &= J^{-1} \xi_z = x_\eta y_\zeta - x_\zeta y_\eta, \\ \hat{\eta}_x &= J^{-1} \eta_x = y_\zeta z_\xi - y_\xi z_\zeta, & \hat{\eta}_y &= J^{-1} \eta_y = z_\zeta x_\xi - z_\xi x_\zeta, & \hat{\eta}_z &= J^{-1} \eta_z = x_\zeta y_\xi - x_\xi y_\zeta, \\ \hat{\zeta}_x &= J^{-1} \zeta_x = y_\xi z_\eta - y_\eta z_\xi, & \hat{\zeta}_y &= J^{-1} \zeta_y = z_\xi x_\eta - z_\eta x_\xi, & \hat{\zeta}_z &= J^{-1} \zeta_z = x_\xi y_\eta - x_\eta y_\xi, \end{aligned} \quad (5)$$

$$\begin{aligned} \hat{\xi}_t &= J^{-1} \xi_t = -x_\tau \hat{\xi}_x - y_\tau \hat{\xi}_y - z_\tau \hat{\xi}_z, \\ \hat{\eta}_t &= J^{-1} \eta_t = -x_\tau \hat{\eta}_x - y_\tau \hat{\eta}_y - z_\tau \hat{\eta}_z, \\ \hat{\zeta}_t &= J^{-1} \zeta_t = -x_\tau \hat{\zeta}_x - y_\tau \hat{\zeta}_y - z_\tau \hat{\zeta}_z. \end{aligned} \quad (6)$$

Although only stationary grids are considered in the present work, Eqs. (6) is still list here for completeness.

If the grids are time-constant, Eq. (3) can be discretized as

$$\delta_\tau Q = -\frac{1}{J^{-1}} \left(\delta_\xi^1 \hat{E} + \delta_\eta^1 \hat{F} + \delta_\zeta^1 \hat{G} \right), \quad (7)$$

where δ^1 is the spatial difference operator acting on the fluxes in Eq. (3) and the subscripts indicate the partial directions, δ_τ represents the time difference operator acting on the dependent variable Q . If δ^1 is the second-order central finite difference scheme and the grid points illuminated in Fig. 1 are used, Eq. (7) can be rewritten as

$$\delta_\tau Q_{i,j,k} = -\frac{1}{J_{i,j,k}^{-1}} \left(\frac{\hat{E}_{i+1,j,k} - \hat{E}_{i-1,j,k}}{2\Delta\xi} + \frac{\hat{F}_{i,j+1,k} - \hat{F}_{i,j-1,k}}{2\Delta\eta} + \frac{\hat{G}_{i,j,k+1} - \hat{G}_{i,j,k-1}}{2\Delta\zeta} \right). \quad (8)$$

In uniform flow field, Q, E, F and G are all constants, Eqs. (7) can be simplified as

$$\delta_\tau Q = -\frac{1}{J^{-1}} \left(I_x^N E + I_y^N F + I_z^N G \right), \quad (9)$$

where

$$\begin{aligned} I_x^N &= \delta_\xi^1 (\hat{\xi}_x) + \delta_\eta^1 (\hat{\eta}_x) + \delta_\zeta^1 (\hat{\zeta}_x), \\ I_y^N &= \delta_\xi^1 (\hat{\xi}_y) + \delta_\eta^1 (\hat{\eta}_y) + \delta_\zeta^1 (\hat{\zeta}_y), \\ I_z^N &= \delta_\xi^1 (\hat{\xi}_z) + \delta_\eta^1 (\hat{\eta}_z) + \delta_\zeta^1 (\hat{\zeta}_z) \end{aligned} \quad (10)$$

and the superscript “N” denotes the numerical values. In uniform flow field, the free-stream preservation property presents as the cancellation of the right hand side of Eq. (9), namely

$$I_x^N = I_y^N = I_z^N = 0. \quad (11)$$

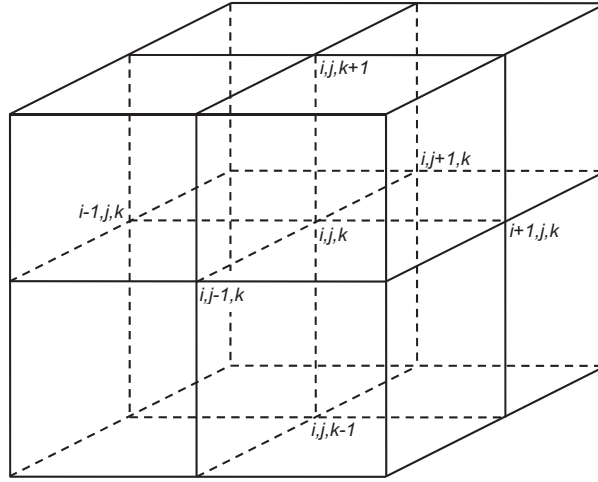


Fig. 1. The grid points in three-dimensional coordinates.

Actually the metrics cancellation of Eq. (11) is the different form of the Surface Conservation Law (SCL), which indicates that the vectorized surfaces should enclose a specific cell volume and must be guaranteed strictly from geometry viewpoint.

If δ^1 is also the second-order central difference scheme, Eqs. (10) can be rewritten as follows.

$$\begin{aligned} I_{xi,j,k}^N &= \frac{\hat{\xi}_{xi+1,j,k} - \hat{\xi}_{xi-1,j,k}}{2\Delta\xi} + \frac{\hat{\eta}_{xi,j,k+1} - \hat{\eta}_{xi,j,k-1}}{2\Delta\eta} + \frac{\hat{\zeta}_{xi,j,k+1} - \hat{\zeta}_{xi,j,k-1}}{2\Delta\zeta}, \\ I_{y,i,j,k}^N &= \frac{\hat{\xi}_{yi+1,j,k} - \hat{\xi}_{yi-1,j,k}}{2\Delta\xi} + \frac{\hat{\eta}_{yi,j,k+1} - \hat{\eta}_{yi,j,k-1}}{2\Delta\eta} + \frac{\hat{\zeta}_{yi,j,k+1} - \hat{\zeta}_{yi,j,k-1}}{2\Delta\zeta}, \\ I_{z,i,j,k}^N &= \frac{\hat{\xi}_{zi+1,j,k} - \hat{\xi}_{zi-1,j,k}}{2\Delta\xi} + \frac{\hat{\eta}_{zi,j,k+1} - \hat{\eta}_{zi,j,k-1}}{2\Delta\eta} + \frac{\hat{\zeta}_{zi,j,k+1} - \hat{\zeta}_{zi,j,k-1}}{2\Delta\zeta}. \end{aligned} \quad (12)$$

By substituting the metrics defined by Eq. (5) into Eq. (12), the SCL terms are not identically zero. This phenomenon was first detected by Steger [2,3]. In order to strictly satisfy the SCL within finite difference frame, many endeavors have been tried, and some of those will be briefly introduced in Section 3.

2.2. Discretizations of governing equations in FVM and SCL

The integral formulation of Eq. (1) in three-dimensional Cartesian coordinates can be gained by using Gauss's divergence theorem.

$$\frac{\partial}{\partial t} \iiint_{\Omega} Q dV + \oint_{\partial\Omega} \vec{F} \cdot d\vec{S} = 0, \quad (13)$$

where the flux vector $\vec{F} = E\vec{i} + F\vec{j} + G\vec{k}$, Ω and $\partial\Omega$ are the whole domain and surface of the super cell respectively illuminated in Fig. 1. On three-dimensional time-constant structured curvilinear grids, Eq. (13) may be discretized by a cell-vertex FVM [13,14]

$$\delta_{\tau} \bar{Q}_{i,j,k} = -\frac{1}{V_{i,j,k}} \begin{pmatrix} \vec{F}_{i+1,j,k} \cdot \vec{\xi}_{i+1,j,k} - \vec{F}_{i-1,j,k} \cdot \vec{\xi}_{i-1,j,k}^V \\ \vec{F}_{i,j,k+1} \cdot \vec{\eta}_{i,j,k+1}^V - \vec{F}_{i,j,k-1} \cdot \vec{\eta}_{i,j,k-1}^V \\ \vec{F}_{i,j,k+1} \cdot \vec{\zeta}_{i,j,k+1}^V - \vec{F}_{i,j,k-1} \cdot \vec{\zeta}_{i,j,k-1}^V \end{pmatrix}, \quad (14)$$

where $\bar{Q}$ is the volumetric averaging variable $\bar{Q} = \frac{1}{V} \iiint_{\Omega} Q dV$, and $\vec{\xi}^V$, $\vec{\eta}^V$ and $\vec{\zeta}^V$ are the vectorized surfaces in ξ , η and ζ directions respectively in FVM. For example, the vectorized surface at point "O" in the ξ direction illustrated in Fig. 2 can be given as

$$\vec{\zeta}_O^V = \vec{S}_{OGDH} + \vec{S}_{OHAE} + \vec{S}_{OEBF} + \vec{S}_{OFCG} = \frac{1}{2} (\vec{AO} \times \vec{EH} + \vec{HG} \times \vec{OD} + \vec{OC} \times \vec{FG} + \vec{EF} \times \vec{BO}). \quad (15)$$

And $V_{i,j,k}$ is the volume of the super cell volume illuminated in Fig. 1, which is formed by the eight cells surrounding the cell vertex (i, j, k) . In FVM the cell volume is enclosed by its vectorized surfaces, and so can be uniquely determined by them. In

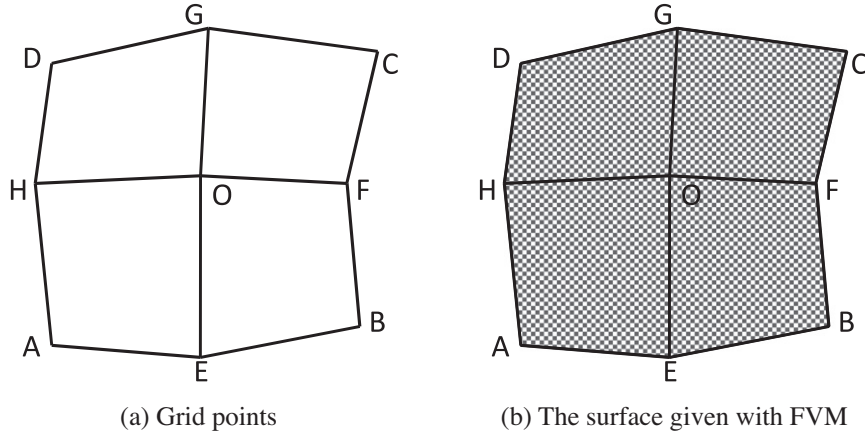


Fig. 2. The vectorized surface at point “O” in ζ direction, the ξ and η directions are the same as the vector $\vec{AE}$ and $\vec{AH}$, respectively, and the ξ , η and ζ directions are components of a right-hand coordinate.

order to correctly carry the information about geometry, the volume of the super cell $V_{ij,k}$ can be calculated by the following expression.

$$V_{ij,k} = V_{ij,k} \left(\vec{\zeta}_{i+1,j,k}^V, \vec{\zeta}_{i-1,j,k}^V, \vec{\eta}_{ij+1,k}^V, \vec{\eta}_{ij-1,k}^V, \vec{\zeta}_{ij,k+1}^V, \vec{\zeta}_{ij,k-1}^V \right).$$

In FVM, in order to calculate the volume of the super cell enclosed by its vectorized surfaces, the super cell is usually partitioned into several pyramids, and to sum all the volumes of pyramids one can get the volume $V_{ij,k}$. As we will see in Fig. 6, an easy way to partition the super cell is to form pyramids with each of the vectorized surfaces calculated by Eq. (15) as bottom surface and with the same apex point “O” in Fig. 6. In such situation, the volume $V_{ij,k}$ can be represented as

$$\begin{aligned} V_{ij,k} &= V_{A_L E_L B_L F_L C_L G_L D_L H_L}^O + V_{A_A L H_L D_L D_R H_R A_R}^O + V_{A_A R E_R B_R B_L E_L A_L}^O + V_{A_R H_R D_R G_R C_R F_R B_R E_R}^O + V_{B_R F_R C_R C_L F_L B_L B}^O + V_{D_D L G_L C_L C_R G_R D_R}^O \\ &= V_{A_L A_R B_L B_R C_L C_R D_L D_R}, \end{aligned} \quad (16)$$

where the subscripts of the uppercase “V” denote the vertex points of the bottom surface of a pyramid and the superscript denotes the apex point of a pyramid, the last volume in Eq. (16) represents the volume as its eight apexes.

The free-stream preservation property of Eq. (13) in uniform flow fields requires

$$\oint_{\partial\Omega} d\vec{S} = 0.$$

which is the integral form of the SCL. The following discretized form of the SCL in FVM can be gotten with the grid points illuminated in Fig. 1

$$\vec{\zeta}_{i+1,j,k}^V - \vec{\zeta}_{i-1,j,k}^V + \vec{\eta}_{ij+1,k}^V - \vec{\eta}_{ij-1,k}^V + \vec{\zeta}_{ij,k+1}^V - \vec{\zeta}_{ij,k-1}^V = 0. \quad (17)$$

The SCL in Eq. (17) is always satisfied profiting from the vectorized surfaces calculation method in Eq. (15), and cell volume V in Eq. (16) is completely enclosed by its corresponding vectorized surfaces. In FVM the vectorized surfaces enclose the cell volume, and the cell volume is uniquely determined by its vectorized surfaces, thus the information about geometry is correctly presented.

3. Discussions on the metrics and Jacobian in FDM from the geometry viewpoint

It is well known that the vectorized surfaces and cell volume within finite volume frame are uniquely determined and correctly carry information about grid geometry, while the metrics and Jacobian within finite difference frame have a variety of equivalent partial-differential expressions, which may have different discretization errors despite that a same difference scheme is applied to discrete the partial derivatives. It can be seen that the right hand sides of Eq. (8) and Eq. (14) are similar to each other, it is naturally expected that some indications might be obtained from FVM to determine the method for calculating the metrics and Jacobian in FDM.

3.1. Discussions on metrics in FDM and vectorized surfaces in FVM

With an eye on the numerical metrics in Eq. (5) from geometry viewpoint, the numerical metrics such as $\hat{\zeta}_x$, $\hat{\zeta}_y$ and $\hat{\zeta}_z$ are supposed to describe the vectorized surface in the ζ direction. However, the numerical metrics calculated by Eq. (5) can not

exactly convey this message of the vectorized surfaces of a cell volume, neither do the conservative form in Refs. [1,6,12]. These two forms of metrics will be further discussed in the following texts of the present subsection.

Considering the grid points illuminated in Fig. 2a, within finite difference frame the vectorized surface at point “O” in the ζ direction can be gained from different metric forms such as the non-conservative form in Eq. (5) and the conservative form in Eq. (21). Within finite volume frame, the vectorized surface at point “O” in the ζ direction is illustrated in Fig. 2b. This vectorized surface is expected to be obtained within finite difference frame and marked as $\vec{S}_{FVM}$.

$$\vec{S}_{FVM} = \vec{\zeta}_O^V = \frac{1}{2} (\vec{AO} \times \vec{EH} + \vec{HG} \times \vec{OD} + \vec{OC} \times \vec{FG} + \vec{EF} \times \vec{BO}). \quad (18)$$

For the grids in Fig. 2a the vectorized surface at point “O” in the ζ direction within finite difference frame can be given from the definition in Eqs. (5).

$$\begin{cases} \hat{\zeta}_x^n = (\delta_\zeta^2 y) (\delta_\eta^2 z) - (\delta_\eta^2 y) (\delta_\zeta^2 z), \\ \hat{\zeta}_y^n = (\delta_\zeta^2 z) (\delta_\eta^2 x) - (\delta_\eta^2 z) (\delta_\zeta^2 x), \\ \hat{\zeta}_z^n = (\delta_\zeta^2 x) (\delta_\eta^2 y) - (\delta_\eta^2 x) (\delta_\zeta^2 y), \end{cases} \quad (19)$$

where the superscript “n” denotes the non-conservative form of metrics and δ^2 is the numerical difference operator in Eqs. (5). If the second-order central difference scheme in all coordinate directions is used as δ^2 , the numerical results of the non-conservative form of metrics in Eq. (19) are

$$\begin{cases} \hat{\zeta}_x^n = \frac{1}{4} [(y_F - y_H)(z_G - z_E) - (z_F - z_H)(y_G - y_E)] = \frac{1}{4} (\vec{HF} \times \vec{EG})_x, \\ \hat{\zeta}_y^n = \frac{1}{4} [(z_F - z_H)(x_G - x_E) - (x_F - x_H)(z_G - z_E)] = \frac{1}{4} (\vec{HF} \times \vec{EG})_y, \\ \hat{\zeta}_z^n = \frac{1}{4} [(x_F - x_H)(y_G - y_E) - (y_F - y_H)(x_G - x_E)] = \frac{1}{4} (\vec{HF} \times \vec{EG})_z, \end{cases} \quad (20)$$

$$\vec{\zeta}^n = \hat{\zeta}_x^n \vec{i} + \hat{\zeta}_y^n \vec{j} + \hat{\zeta}_z^n \vec{k} = \frac{1}{4} \vec{HF} \times \vec{EG} = \frac{1}{2} \vec{S}_{EFGH}.$$

The vectorized surface $\vec{S}_{EFGH}$ in Eq. (20) is illuminated by the black shadow in Fig. 3a. If the vectorized surface defined by Eqs. (20) is used as metrics in Eqs. (12), the SCL cannot be guaranteed due to the abnormality of the vectorized surfaces calculated by the non-conservative metrics of Eqs. (5). At the same time the vectorized surfaces given in Eqs. (20) cannot enclose any cell volume at all, which is one of reasons for the violation of the SCL in FDM.

In order to ensure the SCL with second-order accuracy, Pulliam and Steger [4] presented a special weighted averaging, and the corresponding vectorized surface gained from the averaging procedure is illuminated in Fig. 3b. The same vectorized surface was derived by Vinokur in his review literature [11] within finite volume frame, and his metric method is called as finite-volume-like method in subsequent literatures [1,8]. The substantial reason for the satisfaction of the SCL is that the vectorized surfaces gained from Pulliam and Steger’s averaging procedure can enclose a cell volume despite that the cell volume is more or less different from the control volume enclosed by the surfaces in Eq. (14).

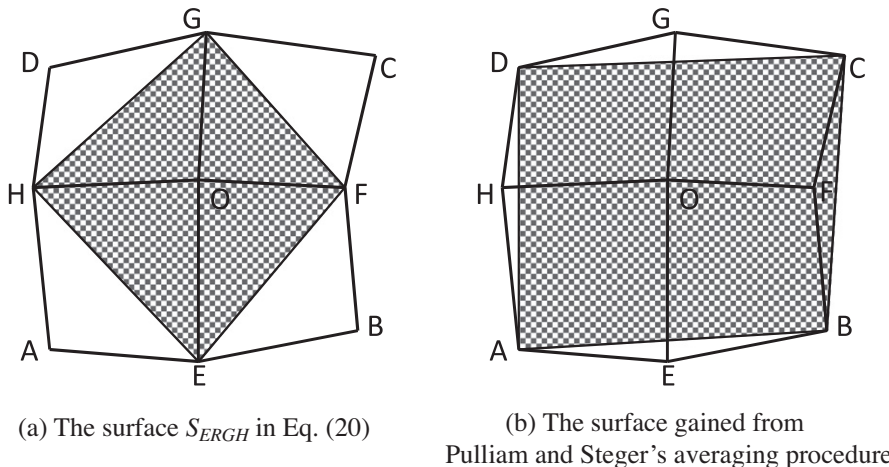


Fig. 3. The surfaces S_{ERGH} in Eq. (20) and Pulliam and Steger’s averaging procedure.

The averaging procedure presented by Pulliam and Steger [4,10] is central type and only second-order accurate. Two momentous shortcomings must be faced, one is the difficulty to implement on boundary and near boundary points where central type numerical operators cannot be applied, and the other is that the averaging procedure is hard to be implemented with high-order difference schemes.

3.1.1. The conservative form of metrics S1

Considering the complexity and hard implementation with high-order difference schemes of the averaging procedure proposed by Pulliam and Steger [4], Thomas and Lombard [6] proposed a conservative form of metrics which has been commended as a milestone in the history of the GCL in FDM and is distinctively marked as S1 in the present work. The terminology “Geometric Conservation Law (GCL)” was also coined by Thomas and Lombard [5,6] who focus on the computation of Jacobian on dynamic grids. The conservative form of metrics S1 may be written as

$$\begin{aligned}\widehat{\xi}_x^{S1} &= (y_\eta z)_\xi - (y_\xi z)_\eta, & \widehat{\xi}_y^{S1} &= (z_\eta x)_\xi - (z_\xi x)_\eta, & \widehat{\xi}_z^{S1} &= (x_\eta y)_\xi - (x_\xi y)_\eta, \\ \widehat{\eta}_x^{S1} &= (y_\xi z)_\xi - (y_\eta z)_\xi, & \widehat{\eta}_y^{S1} &= (z_\xi x)_\xi - (z_\eta x)_\xi, & \widehat{\eta}_z^{S1} &= (x_\xi y)_\xi - (x_\eta y)_\xi, \\ \widehat{\zeta}_x^{S1} &= (y_\xi z)_\eta - (y_\eta z)_\xi, & \widehat{\zeta}_y^{S1} &= (z_\xi x)_\eta - (z_\eta x)_\xi, & \widehat{\zeta}_z^{S1} &= (x_\xi y)_\eta - (x_\eta y)_\xi.\end{aligned}\quad (21)$$

These metrics can be discretized by finite difference operators as

$$\widehat{\xi}_x^{S1} = \delta_\xi^2 (z \delta_\eta^3 y) - \delta_\eta^2 (z \delta_\xi^3 y), \quad \widehat{\xi}_y^{S1} = \delta_\xi^2 (x \delta_\eta^3 z) - \delta_\eta^2 (x \delta_\xi^3 z), \quad \widehat{\xi}_z^{S1} = \delta_\xi^2 (y \delta_\eta^3 x) - \delta_\eta^2 (y \delta_\xi^3 x) \quad (22)$$

and with similar expressions for the other metrics. Hence two-level spatial derivatives are introduced in Eqs. (22), where the outer-level numerical difference operator is marked as δ^2 and the inner-level numerical difference operator is marked as δ^3 . Thomas and Lombard [6] pointed out that if the same second-order central difference operator is used to evaluate all the spatial derivatives in Eq. (21), the SCL terms in Eqs. (12) vanish, namely $I_x^N = I_y^N = I_z^N = 0$.

For high-order FDM, Deng et al. [1] have recently presented a theoretical analysis of the SCL and have derived a sufficient condition to satisfy the SCL based on Thomas and Lombard's conservative form of metrics S1. Deng et al. [1] named the methods satisfying the sufficient condition as the Conservative Metric Method (CMM), which can be exactly presented as: (1) the conservative form of metrics in Eq. (21) is used, and (2) the condition $\delta^1 = \delta^2$ is satisfied. Although δ^3 has no effect on the SCL, the constraint $\delta^3 = \delta^2$ is suggested in Ref. [1], where the extra errors that may be introduced by $\delta^3 \neq \delta^2$ is analyzed. In the present work, the constraint $\delta^3 = \delta^2$ is further discussed from geometry viewpoint by the similarity between metrics in FDM and vectorized surface in FVM.

It can be found out that Thomas and Lombard's [5,6] numerical metrics is a second-order case of the CMM. A high-order case of the CMM for central compact schemes can be found in Visbal and Gaitonde's numerical technique [12]. It is worth to be noted that the CMM can ensure the SCL not only for interior grid points but also for boundary and near boundary grid points. However, if the same difference scheme, such as second-order central difference scheme, is applied to calculate the spatial derivatives in Eqs. (21), it is very difficult to define its corresponding vectorized surface. In order to endow the metrics with geometric information namely the vectorized surfaces in FVM, two other forms of metrics are presented and discussed next.

3.1.2. The conservative form of metrics S2

It is obvious that the metrics defined by the traditional non-conservative form of Eqs. (5) are identical to the metrics defined by the conservative form of Eqs. (21) if

$$(y_\xi)_\eta = (y_\eta)_\xi, \quad (z_\xi)_\eta = (z_\eta)_\xi \quad (23)$$

and with similar conditions for the other cross derivatives. The conditions in the cross derivatives may lead to another conservative form of metrics which is marked as S2.

$$\begin{aligned}\widehat{\xi}_x^{S2} &= (yz_\xi)_\eta - (yz_\eta)_\xi, & \widehat{\xi}_y^{S2} &= (zx_\xi)_\eta - (zx_\eta)_\xi, & \widehat{\xi}_z^{S2} &= (xy_\xi)_\eta - (xy_\eta)_\xi, \\ \widehat{\eta}_x^{S2} &= (yz_\xi)_\xi - (yz_\eta)_\xi, & \widehat{\eta}_y^{S2} &= (zx_\xi)_\xi - (zx_\eta)_\xi, & \widehat{\eta}_z^{S2} &= (xy_\xi)_\xi - (xy_\eta)_\xi, \\ \widehat{\zeta}_x^{S2} &= (yz_\eta)_\xi - (yz_\xi)_\eta, & \widehat{\zeta}_y^{S2} &= (zx_\eta)_\xi - (zx_\xi)_\eta, & \widehat{\zeta}_z^{S2} &= (xy_\eta)_\xi - (xy_\xi)_\eta.\end{aligned}\quad (24)$$

The sufficient condition proved by Deng et al. [1] for the metrics S1 in Eqs. (21) can also satisfy the SCL for the metrics S2 with high-order WCNS, by the reason of the same conservative property of both S1 and S2. However, similar to that of S1, it is also difficult to define the vectorized surfaces for the metrics S2.

3.1.3. The symmetrical conservative form of metrics S3

Before the introduction of the symmetrical conservative metrics, some important properties of linear difference operator δ are given without proof.

$$\delta(c\phi) = c\delta\phi, \quad (25a)$$

$$\delta(\phi + \varphi) = \delta\phi + \delta\varphi, \quad (25b)$$

$$\delta_\xi(\delta_\eta\phi) = \delta_\eta(\delta_\xi\phi). \quad (25c)$$

Here c is an arbitrary constant. The properties of Eq. (25a) and Eq. (25b) can be seen straightly, only the one in Eq. (25c) requires proof, which was given by Deng et al. [1].

Because of the properties of linear difference operator in Eqs. (25), it can be figured out that an arbitrary linear combination of S1 and S2 can satisfy the SCL on the sufficient condition $\delta^1 = \delta^2$ with high-order WCNS. Then it is hard to say which conservative form of the metrics is better than the others from the SCL viewpoint. The following symmetrical conservative form of metrics is derived from the vectorized surface viewpoint and marked as S3.

$$\begin{aligned} \hat{\zeta}_x^{S3} &= \frac{1}{2}(\hat{\zeta}_x^{S1} + \hat{\zeta}_x^{S2}), & \hat{\zeta}_y^{S3} &= \frac{1}{2}(\hat{\zeta}_y^{S1} + \hat{\zeta}_y^{S2}), & \hat{\zeta}_z^{S3} &= \frac{1}{2}(\hat{\zeta}_z^{S1} + \hat{\zeta}_z^{S2}), \\ \hat{\eta}_x^{S3} &= \frac{1}{2}(\hat{\eta}_x^{S1} + \hat{\eta}_x^{S2}), & \hat{\eta}_y^{S3} &= \frac{1}{2}(\hat{\eta}_y^{S1} + \hat{\eta}_y^{S2}), & \hat{\eta}_z^{S3} &= \frac{1}{2}(\hat{\eta}_z^{S1} + \hat{\eta}_z^{S2}), \\ \hat{\varsigma}_x^{S3} &= \frac{1}{2}(\hat{\varsigma}_x^{S1} + \hat{\varsigma}_x^{S2}), & \hat{\varsigma}_y^{S3} &= \frac{1}{2}(\hat{\varsigma}_y^{S1} + \hat{\varsigma}_y^{S2}), & \hat{\varsigma}_z^{S3} &= \frac{1}{2}(\hat{\varsigma}_z^{S1} + \hat{\varsigma}_z^{S2}). \end{aligned} \quad (26)$$

As we will see, if the same second-order central difference scheme in all the coordinate directions is applied to both δ^2 and δ^3 , the numerical result of metrics in the ζ direction in Eqs. (26) can be represented by

$$\vec{\zeta}^{S3} = \frac{1}{8}(\vec{AO} \times \vec{EH} + \vec{HG} \times \vec{OD} + \vec{OC} \times \vec{FG} + \vec{EF} \times \vec{BO}) = \frac{1}{4}\vec{S}_{FVM}. \quad (27)$$

It is very interesting that the numerical result of S3 in Eqs. (26) represents exactly one fourth of the vectorized surface in Fig. 2b, which is derived within finite volume frame. The same conservative form of metrics S3 in Eq. (26) was also proposed by Vinokur and Yee [7] as the coordinate invariant form of metrics, but the geometry meanings were not illuminated.

Some remarks are valuable to be added here as complementarity for the new symmetrical conservative form of metrics S3 in Eqs. (26).

Remark 1. From geometry viewpoint, $\delta^3 = \delta^2$ is a necessary condition for an essential connection or a bridge between the metrics in FDM and the vectorized surfaces in FVM. Then the symmetrical conservative form of metrics S3 with the constraint $\delta^3 = \delta^2$ are suggested to be utilized in FDM.

Remark 2. If δ^2 and δ^3 are the same first-order backward difference scheme in both ξ and η directions, the numerical vectorized surface can be represented in Eq. (28a), and similarly Eq. (28b) represents the result in the situation of the first-order forward difference scheme for both δ^2 and δ^3 in both ξ and η coordinate directions.

$$\vec{\zeta}_b^{S3} = \frac{1}{2}\vec{AO} \times \vec{EH} = \vec{S}_{AE OH}, \quad (28a)$$

$$\vec{\zeta}_f^{S3} = \frac{1}{2}\vec{OC} \times \vec{FG} = \vec{S}_{OF CG}. \quad (28b)$$

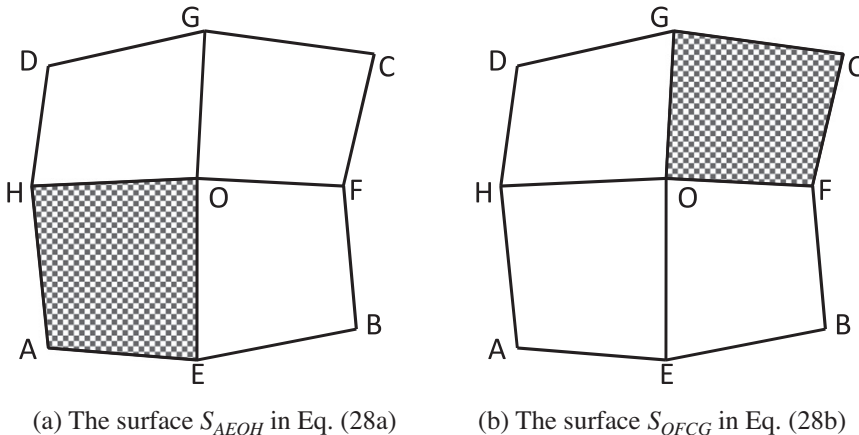


Fig. 4. The surfaces $S_{AE OH}$ and $S_{OF CG}$ in Eq. (28).

The two corresponding vectorized surfaces $\vec{S}_{AEOH}$ and $\vec{S}_{OFCG}$ are illuminated by the black shadow in Fig. 4a and Fig. 4b respectively. Thus if δ^2 and δ^3 are not central type difference scheme, the vectorized surfaces gained with $\delta^3 = \delta^2$ are not central type either.

Remark 3. Similar property as that in Eq. (27) can be obtained by the S3 for the other metrics in the ξ and η directions, and more exactly, the constraint $\delta^3 = \delta^2$ shall be written as

$$\delta_\xi^3 = \delta_\xi^2, \quad \delta_\eta^3 = \delta_\eta^2, \quad \delta_\zeta^3 = \delta_\zeta^2. \quad (29)$$

This constraint can be conveniently realized at boundary and near boundary grid points, where the numerical difference scheme used in one direction is usually different from the ones used in the other directions. For example, if the first-order backward difference scheme is applied in the ξ direction for both δ_ξ^2 and δ_ξ^3 , while the second-order central difference scheme in the η direction for both δ_η^2 and δ_η^3 , the numerical result can be represented in Eq. (30a). Eq. (30b) represents another case that the second-order central difference scheme used in the ξ direction for both δ_ξ^2 and δ_ξ^3 while the first-order forward difference scheme used in the η direction for both δ_η^2 and δ_η^3 .

$$\vec{S}_{bc}^{S3} = \frac{1}{4} (\vec{AO} \times \vec{EH} + \vec{HG} \times \vec{OD}) = \frac{1}{2} \vec{S}_{AEOGDH}, \quad (30a)$$

$$\vec{S}_{cf}^{S3} = \frac{1}{4} (\vec{HG} \times \vec{OD} + \vec{OC} \times \vec{FG}) = \frac{1}{2} \vec{S}_{HOFCGD}. \quad (30b)$$

The two corresponding vectorized surfaces $\vec{S}_{AEOGDH}$ and $\vec{S}_{HOFCGD}$ of Eq. (30a) and Eq. (30b) are illuminated in Fig. 5a and Fig. 5b, respectively. Thus at boundary and near boundary grid points, different numerical schemes can be used in different coordinate directions, but when Eq. (29) is hold, the numerical metrics can represent as vectorized surfaces.

3.2. Discussions on Jacobian in FDM and cell volume in FVM

Similar to the metrics, the Jacobian also has a variety of equivalent partial-differential expressions. From geometry view-point, the proper form of Jacobian is expected to correctly represent the volume of a cell. However, as discussed in the followings, the numerical result of the Jacobian defined by Eq. (4) is hard to convey the message of cell volume.

3.2.1. The traditional form of Jacobian V1

Eq. (4) is a mathematic definition of Jacobian, which is marked as V1 in this paper.

$$J_{V1}^{-1} = x_\xi y_\eta z_\zeta - x_\xi y_\zeta z_\eta + x_\eta y_\zeta z_\xi - x_\eta y_\xi z_\zeta + x_\zeta y_\xi z_\eta - x_\zeta y_\eta z_\xi. \quad (31)$$

The Jacobian can be written in a vector form.

$$J_{V1}^{-1} = (\vec{r}_\xi \times \vec{r}_\eta) \cdot \vec{r}_\zeta, \quad (32)$$

where $\vec{r}$ represents the coordinate vector, and the subscripts indicate the partial derivatives. If the second-order central difference scheme in all coordinate directions is utilized to approximate the derivatives in Eq. (31), and the grids utilized to calculate the Jacobian is illuminated in Fig. 6a, the Jacobian of V1 can be written as

$$8J_O^{-1} = (\vec{O_L O_R} \times \vec{H F}) \cdot \vec{E G} = (\vec{O_L O_R} \times \vec{H F}) \cdot \vec{E O} + (\vec{O_L O_R} \times \vec{H F}) \cdot \vec{O G} = 6(V_{O_L H O_R F}^G + V_{O_L F O_R H}^E).$$

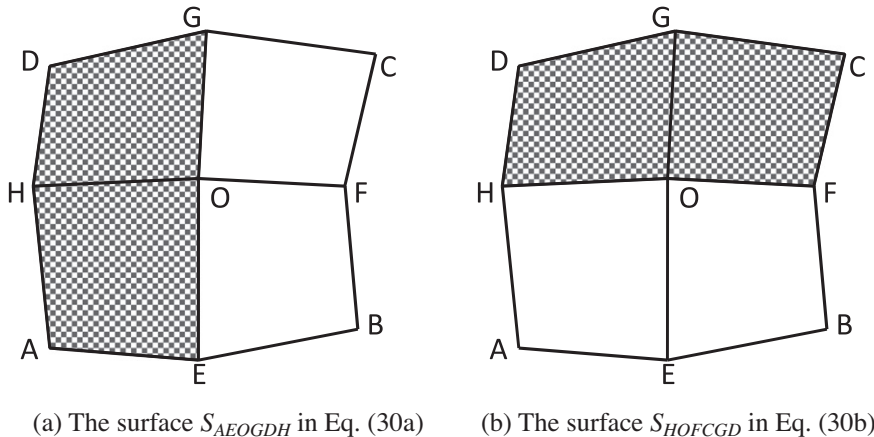
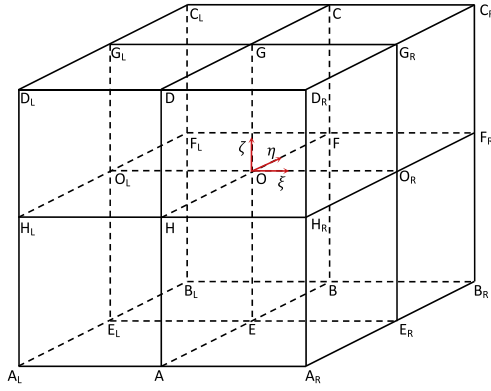
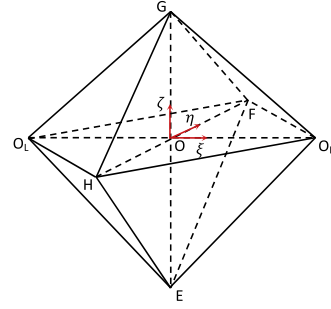
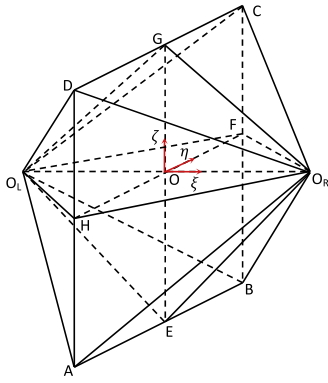
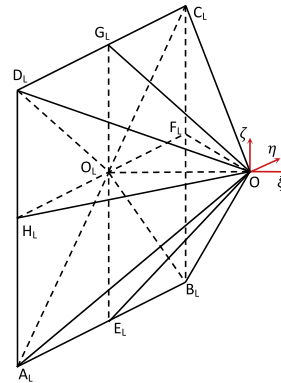


Fig. 5. The surfaces S_{AEOGDH} and S_{HOFCGD} in Eq. (30).



(a) The grid used in Jacobian calculation.

(b) The volume $V_{O_L H O_R F}^G$ and $V_{O_L F O_R H}^E$ in Eq. (33).(c) The volume $V_{A H D G C F B E}^{O_L}$ and $V_{A E B F C G D H}^{O_R}$ in Eq. (38).(d) The volume $V_{A_L E_L B_L F_L C_L G_L D_L H_L}^O$ in Eq. (45).**Fig. 6.** The grids used in Jacobian calculation and the numerical volumes.

namely

$$J_0^{-1} = (\vec{r}_\xi \times \vec{r}_\eta) \cdot \vec{r}_z = \frac{3}{4} (V_{O_L H O_R F}^G + V_{O_L F O_R H}^E). \quad (33)$$

Here the subscripts of the uppercase “V” denote the vertex points of the bottom surface of a pyramid and the superscript denotes the apex point.

The volume $V_{O_L H O_R F}^G$ and $V_{O_L F O_R H}^E$ in Eq. (33) is illuminated in Fig. 6b. Although the shape of this volume is symmetrical, it is not enclosed by the vectorized surfaces of the cell volume illuminated in Fig. 1.

3.2.2. The symmetrical form of Jacobian V2

The Jacobian (volume) can also be expressed as a function of the metrics (vectorized surfaces)

$$\begin{aligned} J^{-1} &= J_{V2}^{-1}(x) = x_\xi \hat{\xi}_x + x_\eta \hat{\eta}_x + x_z \hat{\zeta}_x, \text{ or} \\ J^{-1} &= J_{V2}^{-1}(y) = y_\xi \hat{\xi}_y + y_\eta \hat{\eta}_y + y_z \hat{\zeta}_y, \text{ or} \\ J^{-1} &= J_{V2}^{-1}(z) = z_\xi \hat{\xi}_z + z_\eta \hat{\eta}_z + z_z \hat{\zeta}_z. \end{aligned} \quad (34)$$

Unfortunately, no specific cell volume can be gained by any one of $J_{V2}^{-1}(x)$, $J_{V2}^{-1}(y)$ and $J_{V2}^{-1}(z)$ alone, no matter what kinds of difference schemes and vectorized surfaces are adopted. The three forms of Jacobian in Eqs. (34) can be summed to form a symmetrical Jacobian.

$$J_{V2}^{-1} = \frac{1}{3} (x_\xi \hat{\xi}_x + x_\eta \hat{\eta}_x + x_z \hat{\zeta}_x + y_\xi \hat{\xi}_y + y_\eta \hat{\eta}_y + y_z \hat{\zeta}_y + z_\xi \hat{\xi}_z + z_\eta \hat{\eta}_z + z_z \hat{\zeta}_z). \quad (35)$$

This form of Jacobian is marked as V2. Eq. (35) can be rearranged into a vector form as

$$J_{V2}^{-1} = \frac{1}{3} (\vec{r}_\xi \cdot \vec{\zeta}^D + \vec{r}_\eta \cdot \vec{\eta}^D + \vec{r}_z \cdot \vec{\zeta}^D). \quad (36)$$

where $\vec{\xi}^D$, $\vec{\eta}^D$ and $\vec{\zeta}^D$ are the vectors of metrics in ξ , η and ζ directions respectively in FDM. If the second-order central difference scheme is utilized to approximate the vectors of symmetrical conservative metrics S3 in Eq. (26), similar to Eq. (27) one can get

$$\vec{\xi}^D = \frac{1}{4}\vec{\xi}^V, \quad \vec{\eta}^D = \frac{1}{4}\vec{\eta}^V, \quad \vec{\zeta}^D = \frac{1}{4}\vec{\zeta}^V \quad (37)$$

and so

$$\frac{1}{3}\vec{r}_\xi \cdot \vec{\xi}^D = \frac{1}{6}O_L\vec{O}_R \cdot \vec{\xi}_O^D = \frac{1}{8}\left(V_{AHDGCFBE}^{O_L} + V_{AEBFCGDH}^{O_R}\right). \quad (38)$$

Its corresponding volumes $V_{AHDGCFBE}^{O_L}$ and $V_{AEBFCGDH}^{O_R}$ are illuminated in Fig. 6c. Similarly the other two volumes in Eq. (36) can be gained as

$$\begin{aligned} \frac{1}{3}\vec{r}_\eta \cdot \vec{\eta}^D &= \frac{1}{6}\vec{H}\vec{F} \cdot \vec{\eta}_O^D = \frac{1}{8}\left(V_{GGLOLEEE_ROR_G}^H + V_{GGROREEE_LOL_G}^F\right), \\ \frac{1}{3}\vec{r}_\zeta \cdot \vec{\zeta}^D &= \frac{1}{6}\vec{E}\vec{G} \cdot \vec{\zeta}_O^D = \frac{1}{8}\left(V_{HHLOLEFF_ROR_H}^E + V_{HHROREFF_LOL_H}^G\right) \end{aligned}$$

and the volume calculated by Eq. (35) can be written as

$$J_0^{-1} = \frac{1}{3}\left(\vec{r}_\xi \cdot \vec{\xi}^D + \vec{r}_\eta \cdot \vec{\eta}^D + \vec{r}_\zeta \cdot \vec{\zeta}^D\right) = \frac{1}{8}\left(V_{AHDGCFBE}^{O_L} + V_{GGLOLEEE_ROR_G}^H + V_{HHLOLEFF_ROR_H}^E + V_{AEBFCGDH}^{O_R} + V_{GGROREEE_LOL_G}^F + V_{HHROREFF_LOL_H}^G\right).$$

However, these volumes are still irregular and cannot yet fulfill the control volume enclosed by the vectorized surfaces (metrics) due to its non-conservative form in Eq. (35). In order to get a volume-like Jacobian, a new symmetrical conservative Jacobian will be derived in the next subsection.

3.2.3. The symmetrical conservative form of Jacobian V3

Let us consider a conservative form of Jacobian, which can be obtained based on Eqs. (34) and written as

$$\begin{aligned} J^{-1} &= J_{V3}^{-1}(x) = \left(\widehat{x\xi_x}\right)_\xi + \left(\widehat{x\eta_x}\right)_\eta + \left(\widehat{x\zeta_x}\right)_\zeta - xI_x, \text{ or} \\ J^{-1} &= J_{V3}^{-1}(y) = \left(\widehat{y\xi_y}\right)_\xi + \left(\widehat{y\eta_y}\right)_\eta + \left(\widehat{y\zeta_y}\right)_\zeta - yI_y, \text{ or} \\ J^{-1} &= J_{V3}^{-1}(z) = \left(\widehat{z\xi_z}\right)_\xi + \left(\widehat{z\eta_z}\right)_\eta + \left(\widehat{z\zeta_z}\right)_\zeta - zI_z, \end{aligned} \quad (39)$$

where I_x , I_y and I_z are exactly the SCL terms. If the SCL is satisfied exactly, namely $I_x = I_y = I_z = 0$, Eq. (39) may be simplified to the following completely conservative forms

$$\begin{aligned} J_{V3}^{-1}(x) &= \left(\widehat{x\xi_x}\right)_\xi + \left(\widehat{x\eta_x}\right)_\eta + \left(\widehat{x\zeta_x}\right)_\zeta, \text{ or} \\ J_{V3}^{-1}(y) &= \left(\widehat{y\xi_y}\right)_\xi + \left(\widehat{y\eta_y}\right)_\eta + \left(\widehat{y\zeta_y}\right)_\zeta, \text{ or} \\ J_{V3}^{-1}(z) &= \left(\widehat{z\xi_z}\right)_\xi + \left(\widehat{z\eta_z}\right)_\eta + \left(\widehat{z\zeta_z}\right)_\zeta. \end{aligned} \quad (40)$$

Similar to that of V2 no specific cell volume can be gained by each one of $J_{V3}^{-1}(x)$, $J_{V3}^{-1}(y)$ and $J_{V3}^{-1}(z)$ alone. The three forms of Jacobian in Eqs. (40) can also be summed to form a symmetrical conservative form as Eq. (41), and the form is marked as V3.

$$J_{V3}^{-1} = \frac{1}{3}\left[\left(\widehat{x\xi_x} + \widehat{y\xi_y} + \widehat{z\xi_z}\right)_\xi + \left(\widehat{x\eta_x} + \widehat{y\eta_y} + \widehat{z\eta_z}\right)_\eta + \left(\widehat{x\zeta_x} + \widehat{y\zeta_y} + \widehat{z\zeta_z}\right)_\zeta\right] = \frac{1}{3}\left[\left(\vec{r} \cdot \vec{\xi}^D\right)_\xi + \left(\vec{r} \cdot \vec{\eta}^D\right)_\eta + \left(\vec{r} \cdot \vec{\zeta}^D\right)_\zeta\right]. \quad (41)$$

With the second-order central difference scheme, Eq. (41) may be written as

$$6J_{ij,k}^{-1} = \vec{r}_{i+1,j,k} \cdot \vec{\xi}_{i+1,j,k}^D - \vec{r}_{i-1,j,k} \cdot \vec{\xi}_{i-1,j,k}^D + \vec{r}_{i,j,k+1} \cdot \vec{\eta}_{i,j,k+1}^D - \vec{r}_{i,j,k-1} \cdot \vec{\eta}_{i,j,k-1}^D + \vec{r}_{i,j,k+1} \cdot \vec{\zeta}_{i,j,k+1}^D - \vec{r}_{i,j,k-1} \cdot \vec{\zeta}_{i,j,k-1}^D. \quad (42)$$

The metrics calculated by the S3 with proper finite difference schemes can exactly express the vectorized surfaces as well as ensure the SCL, i.e.

$$\vec{\xi}_{i+1,j,k}^D - \vec{\xi}_{i-1,j,k}^D + \vec{\eta}_{i,j,k+1}^D - \vec{\eta}_{i,j,k-1}^D + \vec{\zeta}_{i,j,k+1}^D - \vec{\zeta}_{i,j,k-1}^D = 0. \quad (43)$$

Then we can get

$$\begin{aligned} 6J_{ij,k}^{-1} &= (\vec{r}_{i+1,j,k} - \vec{r}_{i,j,k}) \cdot \vec{\xi}_{i+1,j,k}^D + (\vec{r}_{i,j,k} - \vec{r}_{i-1,j,k}) \cdot \vec{\xi}_{i-1,j,k}^D + (\vec{r}_{i,j,k+1} - \vec{r}_{i,j,k}) \cdot \vec{\eta}_{i,j,k+1}^D + (\vec{r}_{i,j,k} - \vec{r}_{i,j,k-1}) \cdot \vec{\eta}_{i,j,k-1}^D \\ &\quad + (\vec{r}_{i,j,k+1} - \vec{r}_{i,j,k}) \cdot \vec{\zeta}_{i,j,k+1}^D + (\vec{r}_{i,j,k} - \vec{r}_{i,j,k-1}) \cdot \vec{\zeta}_{i,j,k-1}^D, \end{aligned} \quad (44)$$

where

$$(\vec{r}_{i,j,k} - \vec{r}_{i-1,j,k}) \cdot \vec{\xi}_{i-1,j,k}^D = \frac{1}{8} \left[\begin{aligned} &\vec{O_L O} \cdot (\vec{A_L E_L} \times \vec{E_L O_L}) + \vec{O_L O} \cdot (\vec{A_L O_L} \times \vec{O_L H_L}) + \\ &\vec{O_L O} \cdot (\vec{H_L O_L} \times \vec{O_L G_L}) + \vec{O_L O} \cdot (\vec{H_L G_L} \times \vec{G_L D_L}) + \\ &\vec{O_L O} \cdot (\vec{O_L F_L} \times \vec{F_L C_L}) + \vec{O_L O} \cdot (\vec{O_L C_L} \times \vec{C_L G_L}) + \\ &\vec{O_L O} \cdot (\vec{E_L B_L} \times \vec{B_L F_L}) + \vec{O_L O} \cdot (\vec{E_L F_L} \times \vec{F_L O_L}), \end{aligned} \right] = \frac{3}{4} V_{A_L E_L B_L F_L C_L G_L D_L H_L}^0. \quad (45)$$

The corresponding volume $V_{A_L E_L B_L F_L C_L G_L D_L H_L}^0$ in Eq. (45) is illuminated in Fig. 6d. Analogously, the other volumes in Eq. (44) can be represented as follows.

$$\begin{aligned} (\vec{r}_{i+1,j,k} - \vec{r}_{i,j,k}) \cdot \vec{\xi}_{i+1,j,k}^D &= \frac{3}{4} V_{A_R H_R D_R C_R G_R F_R B_R E_R}^0, \\ (\vec{r}_{i,j,k} - \vec{r}_{i,j-1,k}) \cdot \vec{\eta}_{i,j-1,k}^D &= \frac{3}{4} V_{A_L H_L D_L D_R H_R A_R}^0, \\ (\vec{r}_{i,j+1,k} - \vec{r}_{i,j,k}) \cdot \vec{\eta}_{i,j+1,k}^D &= \frac{3}{4} V_{B_R F_R C_R C_L F_L B_L}^0, \\ (\vec{r}_{i,j,k} - \vec{r}_{i,j,k-1}) \cdot \vec{\xi}_{i,j,k-1}^D &= \frac{3}{4} V_{A_R E_R B_R B_L E_L A_L}^0, \\ (\vec{r}_{i,j,k+1} - \vec{r}_{i,j,k}) \cdot \vec{\xi}_{i,j,k+1}^D &= \frac{3}{4} V_{D_D L G_L C_L C_R G_R D_R}^0. \end{aligned}$$

So the volume calculated by the V3 can be got by gathering all the volume parts together.

$$J_0^{-1} = \frac{1}{3} \left[(\vec{r} \cdot \vec{\xi}^D)_\xi + (\vec{r} \cdot \vec{\eta}^D)_\eta + (\vec{r} \cdot \vec{\xi}^D)_\zeta \right] = \frac{1}{8} V_{A_L A_R B_L B_R C_L C_R D_L D_R}. \quad (46)$$

Thus the volume calculated by the symmetrical conservative form of Jacobian V3 is exactly one eighth of the cell volume of FVM in Eq. (16).

It is well known that, in FVM, a cell volume is always enclosed by its vectorized surfaces and simultaneously bounded by these surfaces, in other words, the volume should be uniquely determined by its vectorized surfaces. Interestingly, in FDM, the symmetrical conservative form of Jacobian V3 is just enclosed by its vectorized surfaces if the spatial derivatives in Eqs. (26) and (41) are all calculated by the same second-order central difference scheme. Then V3 in Eq. (41) is suggested to be a preferable form of Jacobian from the geometry viewpoint.

If all the spatial derivatives are discretized by the second-order central difference scheme, the Jacobian V1, V2 and V3 would involve different grid points and the corresponding numerical values generally may be different. But when all the quadrilaterals in the grid are parallelograms, the Jacobian V1, V2 and V3 will be equivalent. The differences between V1, V2 and V3 exactly represent the deviation of quadrilaterals from parallelograms, thus the relative error of V1, V2 and V3 can be somewhat devised to detect the quality of the grid, such as

$$V_M = \max(V^{12}, V^{23}, V^{13}), \quad V^{12} = \frac{|V^1 - V^2|}{V^2}, \quad V^{23} = \frac{|V^2 - V^3|}{V^3}, \quad V^{13} = \frac{|V^1 - V^3|}{V^3}.$$

For the grid of high quality, V_M is small, otherwise V_M is relatively large.

3.3. The derivation of Symmetrical Conservative Metric Method (SCMM)

By substituting the vectorized surfaces in Eq. (27) and the Jacobian in Eq. (46) into the discretized governing equations, namely Eq. (8), we can see that the physical variable $\delta_\tau Q$ calculated by the FDM with second-order accuracy is actually similar to that of Eq. (14) which is discretized by the second-order FVM. In other words, if the metrics are calculated with the symmetrical conservative form of S3 and Jacobian is computed with the symmetrical conservative form of V3, the same numerical results would be achieved for both FDM and FVM with the same second-order accuracy.

Though this similarity only exists when second-order central scheme is used, it give us an inspiration that we may get a general method to calculate the geometry metrics and Jacobian for high-order finite difference schemes. In this paper, this method is named as Symmetrical Conservative Metric Method (SCMM) which can be exactly defined by:

- (1) the symmetrical conservative form of metrics S3 in Eqs. (26) is adopted,
- (2) the symmetrical conservative form of Jacobian V3 in Eq. (41) is adopted, and
- (3) the numerical derivative operators δ^1 to calculate the flux derivatives in Eq. (3) and the spatial derivatives in Eq. (41), the numerical derivative operators δ^2 to calculate the outer-level derivatives of metrics in Eqs. (26) and the numerical derivative operators δ^3 to calculate the inner-level derivatives of metrics in Eqs. (26) have to be all the same difference scheme in the same coordinate direction, i.e.

$$\delta_\xi^1 = \delta_\xi^2 = \delta_\xi^3, \quad \delta_\eta^1 = \delta_\eta^2 = \delta_\eta^3, \quad \delta_\zeta^1 = \delta_\zeta^2 = \delta_\zeta^3. \quad (47)$$

4. Implementation to high-order finite difference procedure

The Symmetrical Conservative Metric Method (SCMM) proposed in previous section needs the symmetrical conservative metrics and Jacobian, which may be rewritten completely as follows,

Symmetrical conservative metrics S3

$$\begin{cases} \widehat{\zeta}_x^{S3} = \frac{1}{2} \left[(zy_\eta)_\zeta + (yz_\zeta)_\eta - (zy_\zeta)_\eta - (yz_\eta)_\zeta \right], \\ \widehat{\zeta}_y^{S3} = \frac{1}{2} \left[(xz_\eta)_\zeta + (zx_\zeta)_\eta - (xz_\zeta)_\eta - (zx_\eta)_\zeta \right], \\ \widehat{\zeta}_z^{S3} = \frac{1}{2} \left[(yx_\eta)_\zeta + (xy_\zeta)_\eta - (yx_\zeta)_\eta - (xy_\eta)_\zeta \right]. \\ \widehat{\eta}_x^{S3} = \frac{1}{2} \left[(zy_\zeta)_\xi + (yz_\xi)_\zeta - (zy_\xi)_\zeta - (yz_\zeta)_\xi \right], \\ \widehat{\eta}_y^{S3} = \frac{1}{2} \left[(xz_\xi)_\zeta + (zx_\zeta)_\xi - (xz_\zeta)_\xi - (zx_\xi)_\zeta \right], \\ \widehat{\eta}_z^{S3} = \frac{1}{2} \left[(yx_\zeta)_\xi + (xy_\xi)_\zeta - (yx_\xi)_\zeta - (xy_\zeta)_\xi \right]. \\ \widehat{\zeta}_x^{S3} = \frac{1}{2} \left[(zy_\xi)_\eta + (yz_\eta)_\xi - (zy_\eta)_\xi - (yz_\xi)_\eta \right], \\ \widehat{\zeta}_y^{S3} = \frac{1}{2} \left[(xz_\xi)_\eta + (zx_\eta)_\xi - (xz_\eta)_\xi - (zx_\xi)_\eta \right], \\ \widehat{\zeta}_z^{S3} = \frac{1}{2} \left[(yx_\xi)_\eta + (xy_\eta)_\xi - (yx_\eta)_\xi - (xy_\xi)_\eta \right]. \end{cases} \quad (48)$$

The delta forms of S3 in Eqs. (48) are

$$\begin{cases} \widehat{\zeta}_x^{S3} = \frac{1}{2} \left[\delta_\zeta^2 (z\delta_\eta^3 y) + \delta_\eta^2 (y\delta_\zeta^3 z) - \delta_\eta^2 (z\delta_\zeta^3 y) - \delta_\zeta^2 (y\delta_\eta^3 z) \right], \\ \widehat{\zeta}_y^{S3} = \frac{1}{2} \left[\delta_\zeta^2 (x\delta_\eta^3 z) + \delta_\eta^2 (z\delta_\zeta^3 x) - \delta_\eta^2 (x\delta_\zeta^3 z) - \delta_\zeta^2 (z\delta_\eta^3 x) \right], \\ \widehat{\zeta}_z^{S3} = \frac{1}{2} \left[\delta_\zeta^2 (y\delta_\eta^3 x) + \delta_\eta^2 (x\delta_\zeta^3 y) - \delta_\eta^2 (y\delta_\zeta^3 x) - \delta_\zeta^2 (x\delta_\eta^3 y) \right]. \\ \widehat{\eta}_x^{S3} = \frac{1}{2} \left[\delta_\xi^2 (z\delta_\zeta^3 y) + \delta_\zeta^2 (y\delta_\xi^3 z) - \delta_\zeta^2 (z\delta_\xi^3 y) - \delta_\xi^2 (y\delta_\zeta^3 z) \right], \\ \widehat{\eta}_y^{S3} = \frac{1}{2} \left[\delta_\xi^2 (x\delta_\zeta^3 z) + \delta_\zeta^2 (z\delta_\xi^3 x) - \delta_\zeta^2 (x\delta_\xi^3 z) - \delta_\xi^2 (z\delta_\zeta^3 x) \right], \\ \widehat{\eta}_z^{S3} = \frac{1}{2} \left[\delta_\xi^2 (y\delta_\zeta^3 x) + \delta_\zeta^2 (x\delta_\xi^3 y) - \delta_\zeta^2 (y\delta_\xi^3 x) - \delta_\xi^2 (x\delta_\zeta^3 y) \right]. \\ \widehat{\zeta}_x^{S3} = \frac{1}{2} \left[\delta_\eta^2 (z\delta_\xi^3 y) + \delta_\xi^2 (y\delta_\eta^3 z) - \delta_\xi^2 (z\delta_\eta^3 y) - \delta_\eta^2 (y\delta_\xi^3 z) \right], \\ \widehat{\zeta}_y^{S3} = \frac{1}{2} \left[\delta_\eta^2 (x\delta_\xi^3 z) + \delta_\xi^2 (z\delta_\eta^3 x) - \delta_\xi^2 (x\delta_\eta^3 z) - \delta_\eta^2 (z\delta_\xi^3 x) \right], \\ \widehat{\zeta}_z^{S3} = \frac{1}{2} \left[\delta_\eta^2 (y\delta_\xi^3 x) + \delta_\xi^2 (x\delta_\eta^3 y) - \delta_\xi^2 (y\delta_\eta^3 x) - \delta_\eta^2 (x\delta_\xi^3 y) \right]. \end{cases} \quad (49)$$

Symmetrical conservative form of Jacobian V3

$$J_{V3}^{-1} = \frac{1}{3} \left[(x\widehat{\zeta}_x + y\widehat{\zeta}_y + z\widehat{\zeta}_z)_\xi + (x\widehat{\eta}_x + y\widehat{\eta}_y + z\widehat{\eta}_z)_\eta + (x\widehat{\zeta}_x + y\widehat{\zeta}_y + z\widehat{\zeta}_z)_\zeta \right] = \frac{1}{3} \left[(\vec{r} \cdot \vec{\zeta})_\xi + (\vec{r} \cdot \vec{\eta})_\eta + (\vec{r} \cdot \vec{\zeta})_\zeta \right] \quad (50)$$

and its delta form can be described as:

$$J_{V3}^{-1} = \frac{1}{3} \left[\delta_\xi^1 (x\widehat{\zeta}_x + y\widehat{\zeta}_y + z\widehat{\zeta}_z) + \delta_\eta^1 (x\widehat{\eta}_x + y\widehat{\eta}_y + z\widehat{\eta}_z) + \delta_\zeta^1 (x\widehat{\zeta}_x + y\widehat{\zeta}_y + z\widehat{\zeta}_z) \right] = \frac{1}{3} \left[\delta_\xi^1 (\vec{r} \cdot \vec{\zeta}) + \delta_\eta^1 (\vec{r} \cdot \vec{\eta}) + \delta_\zeta^1 (\vec{r} \cdot \vec{\zeta}) \right]. \quad (51)$$

The implementation of the SCMM on high-order linear schemes are trivial, where the only requirement is that the same difference scheme is applied for all the spatial derivatives in Eqs. (3), (48) and (50). However, some special cares shall be paid on other kind of implementation, and a sixth-order WCNS [15] is chosen as an example to show the procedures on implementing the SCMM on WCNS and MUSCL-type¹ schemes.

With the concept of WCNS proposed by Deng and Zhang [15], three steps would be taken to calculate the spatial derivatives in Eq. (3). The first step is interpolation, which means to interpolate the metrics and the flow-field variables to cell edges from the values at cell nodes, usually different schemes, which satisfy the accuracy demand, may be used in this step, generally the

¹ MUSCL: Monotonic Upstream-centered Schemes for Conservation Laws [19].

interpolating schemes used for metrics shall be central type, while the ones for the flow-field variables may be upwind biased or even nonlinear. The second step is calculating the fluxes at cell edges, in this paper only the standard Roe flux [16] is applied. The last step is calculating the cell-node spatial derivatives from the cell-edge fluxes, and the numerical schemes applied in this step are usually central type. By inserting the interpolated cell-edge values in the first step into the numerical spatial derivatives in the last step, one can obtain the virtual numerical spatial derivative operators acting on the metrics, and the interpolation method for flow-field variables has no effect on the SCL, neither dose the flux splitting method used in the second step.

Let us only consider the ξ direction without loss of generality, the sixth-order WCNS may be expressed as

$$\frac{\partial \hat{E}_i}{\partial \xi} = \frac{75}{64\Delta\xi} (\hat{E}_{i+1/2} - \hat{E}_{i-1/2}) - \frac{25}{384\Delta\xi} (\hat{E}_{i+3/2} - \hat{E}_{i-3/2}) + \frac{3}{640\Delta\xi} (\hat{E}_{i+5/2} - \hat{E}_{i-5/2}), \quad (52)$$

where

$$\hat{E}_{i+1/2} = \hat{E} \left(U_{i+1/2}^L, U_{i+1/2}^R, \hat{\xi}_{x,i+1/2}, \hat{\xi}_{y,i+1/2}, \hat{\xi}_{z,i+1/2} \right) \quad (53)$$

is the cell-edge flux and calculated by the Roe flux method [16] in this paper. The cell-edge metrics in Eq. (53) are calculated from cell-node metrics by the following sixth-order interpolation

$$\hat{\xi}_{x,i+1/2} = \frac{150}{256} (\hat{\xi}_{x,i+1} + \hat{\xi}_{x,i}) - \frac{25}{256} (\hat{\xi}_{x,i+2} + \hat{\xi}_{x,i-1}) + \frac{3}{256} (\hat{\xi}_{x,i+3} + \hat{\xi}_{x,i-2}) \quad (54)$$

and with similar relations for the other metrics.

In order to satisfy the SCL as well as to endow the metrics and the Jacobian with their corresponding geometry information as vectorized surfaces and cell volume, respectively, the symmetrical conservative form of metrics S3 in Eqs. (26), the symmetrical conservative form of Jacobian V3 in Eq. (50), and the constraints on the difference operators $\delta^{1,2,3}$ shall all be followed on calculating the metrics and Jacobian for the WCNS.

To summarize the procedure on implementing the SCMM to the WCNS, the difference operators for the metrics are

$$\delta^1 = \delta^2 = \delta^3 : \begin{cases} a_{i+1/2} = \frac{150}{256} (a_{i+1} + a_i) - \frac{25}{256} (a_{i+2} + a_{i-1}) + \frac{3}{256} (a_{i+3} + a_{i-2}), \\ \frac{\partial a_i}{\partial \xi} = \frac{75}{64\Delta\xi} (a_{i+1/2} - a_{i-1/2}) - \frac{25}{384\Delta\xi} (a_{i+3/2} - a_{i-3/2}) + \frac{3}{640\Delta\xi} (a_{i+5/2} - a_{i-5/2}) \end{cases} \quad (55)$$

and the spatial derivatives in Eqs. (21, 24, 26, 41) are all calculated by the difference operator defined by Eqs. (55) except boundary and near boundary points where different operators may be applied in different coordinate directions provided that $\delta_\xi^1 = \delta_\xi^2 = \delta_\xi^3$, $\delta_\eta^1 = \delta_\eta^2 = \delta_\eta^3$ and $\delta_\zeta^1 = \delta_\zeta^2 = \delta_\zeta^3$.

It should be pointed out that the methods of calculating the cell-edge flow-field variables $U_{i\pm 1/2}$ in Eqs. (7) have no effects on the SCL, then any upwind-based linear or nonlinear interpolation/reconstruction methods, such as that in the WCNS and the MUSCL-type schemes, may be used to calculate $U_{i+1/2}^L$ and $U_{i+1/2}^R$. In this paper, $U_{i+1/2}^L$ and $U_{i+1/2}^R$ are calculated by the optimal version of the fifth-order nonlinear weighted interpolation proposed by Deng and Zhang [15].

$$U_{i+1/2}^L = \frac{1}{128} (+3U_{i-2} - 20U_{i-1} + 90U_i + 60U_{i+1} - 5U_{i+2}),$$

$$U_{i+1/2}^R = \frac{1}{128} (-5U_{i-1} + 60U_i + 90U_{i+1} - 20U_{i+2} + 3U_{i+3}).$$

5. Numerical tests

In this section, the numerical tests including a linear wave propagation problem and an Euler isentropic vortex moving problem in randomized grids with different perturbation amplitudes are conducted in order to investigate the influence of the above forms of metrics and Jacobian on the computational results. Finally, the three dimensional flow around the typical ONERA M6 wing is calculated.

5.1. Linear wave propagation problems

The linear wave propagation problem is chosen to clearly investigate the property of different forms of metrics and Jacobian. The governing equation in Cartesian coordinates is

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} + b \frac{\partial u}{\partial y} + c \frac{\partial u}{\partial z} = 0, \quad (56)$$

where a , b and c are all constants.

In three-dimensional curvilinear coordinates, Eq. (56) may be written as

$$\frac{\partial \hat{Q}}{\partial \tau} + \frac{\partial \hat{E}}{\partial \xi} + \frac{\partial \hat{F}}{\partial \eta} + \frac{\partial \hat{G}}{\partial \zeta} = 0, \quad (57)$$

where

$$\begin{aligned}\hat{Q} &= J^{-1}u, \\ \hat{E} &= \hat{\zeta}_t u + \hat{\zeta}_x a u + \hat{\zeta}_y b u + \hat{\zeta}_z c u, \\ \hat{F} &= \hat{\eta}_t u + \hat{\eta}_x a u + \hat{\eta}_y b u + \hat{\eta}_z c u, \\ \hat{G} &= \hat{\zeta}_t u + \hat{\zeta}_x a u + \hat{\zeta}_y b u + \hat{\zeta}_z c u.\end{aligned}$$

In this paper the constants are set to $a = 4, b = 2, c = 1$.

A nominal $51 \times 51 \times 51$ uniform mesh is generated in the domain $-1 \leq x \leq 1, -1 \leq y \leq 1$ and $-1 \leq z \leq 1$. The interior grids are then perturbed by random disturbances with different amplitudes A of the grid size in each coordinate direction, while the boundary points on the edges of the computational field are left unperturbed for easy boundary treatment. The grids illuminated in Fig. 7 is the sub-zone with $12 \leq i \leq 40, 12 \leq j \leq 40$ and $k = 26$. All the boundaries are treated to be

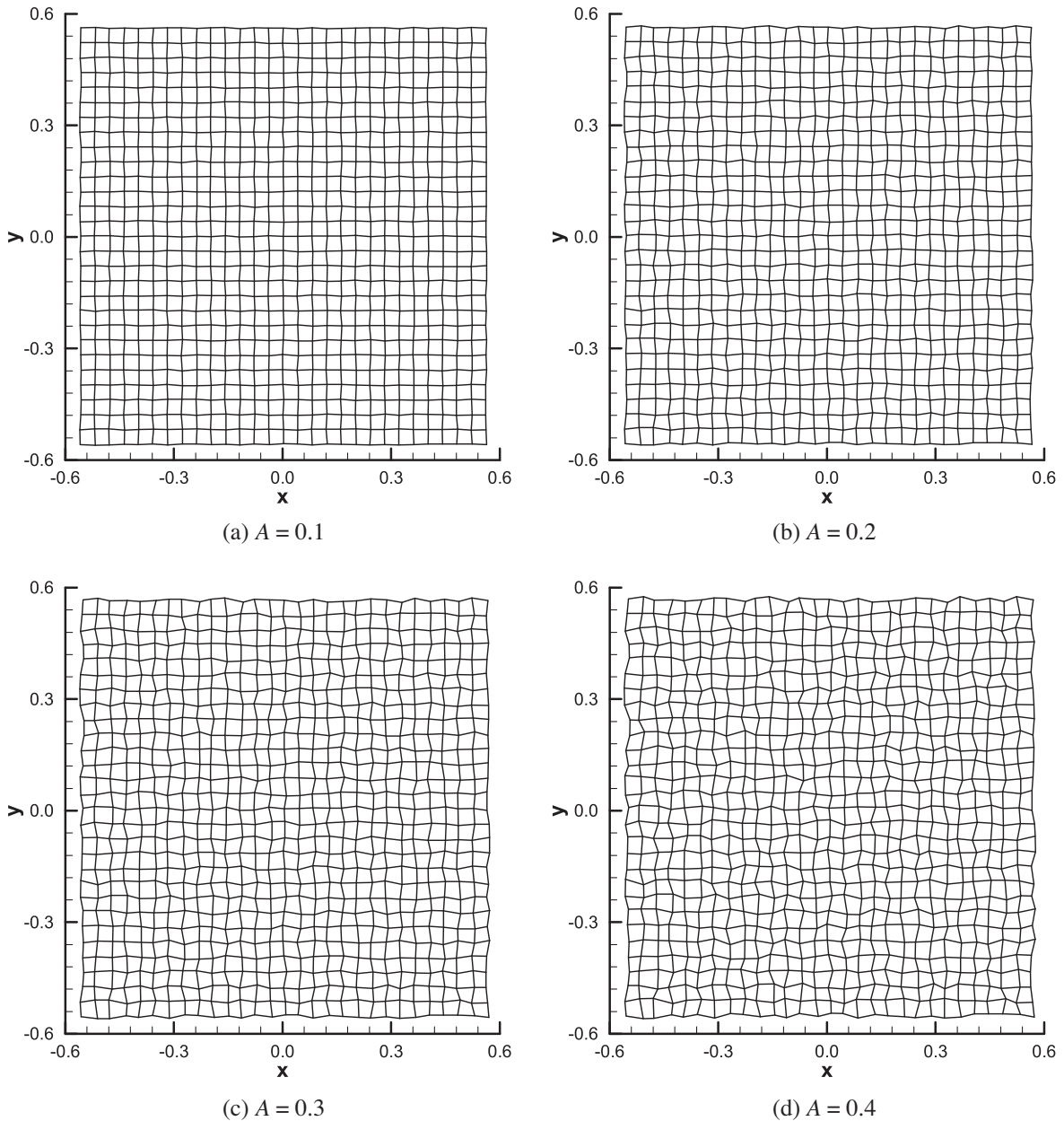


Fig. 7. Randomized grids with different perturbation coefficients A .

periodical for flow-field variables, metrics and Jacobian. The third-order total variation diminishing Runge–Kutta scheme presented by Shu and Osher [17] is adopted for time integration, and the non-dimensional time step $\Delta t = 0.001$ is used for all the numerical tests on this grid in the present work. The computations are carried out until time $t = 2$, at which moment the flow fields move back to its original position of the initial conditions.

The initial condition is

$$u_0 = \begin{cases} \sin[\pi(x + 0.5)]\sin[\pi(y + 0.5)]\sin[\pi(z + 0.5)] & \text{if } (x, y, z) \in [-0.5, 0.5], \\ 0 & \text{else.} \end{cases} \quad (58)$$

The numerical results of the linear wave propagation problem with sixth-order accuracy procedure are shown in Table 1–4. The maximum value of $|I_x^N|$, $|I_y^N|$ and $|I_z^N|$ which are calculated by three different metrics of S1, S2 and S3 are listed in these tables, and the L_1 , L_2 and L_∞ error norms of u between the numerical results and the analytic solutions are included also. The numerical results of I_x^N , I_y^N and I_z^N are all machine zero, while it is interesting that the numerical errors of metrics

Table 1
Errors of u for linear wave propagation problem in randomized grid with $A = 0.1$.

$A = 0.1$		S1	S2	S3
$\max(I_x^N , I_y^N , I_z^N)$		0.84184E–15	0.74457E–15	0.43117E–15
L_1	V1	0.15167E–02	0.15047E–02	0.14152E–02
	V2	0.14075E–02	0.14015E–02	0.13636E–02
	V3	0.13937E–02	0.13968E–02	0.13588E–02
L_2	V1	0.41481E–02	0.41065E–02	0.38022E–02
	V2	0.37496E–02	0.37333E–02	0.35656E–02
	V3	0.36818E–02	0.36866E–02	0.35168E–02
L_∞	V1	0.48627E–01	0.46782E–01	0.41781E–01
	V2	0.39592E–01	0.39562E–01	0.36934E–01
	V3	0.37278E–01	0.36999E–01	0.34034E–01

Table 2
Errors of u for linear wave propagation problem in randomized grid with $A = 0.2$.

$A = 0.2$		S1	S2	S3
$\max(I_x^N , I_y^N , I_z^N)$		0.86827E–15	0.93648E–15	0.68993E–15
L_1	V1	0.25907E–02	0.25396E–02	0.20464E–02
	V2	0.18583E–02	0.18491E–02	0.15585E–02
	V3	0.17471E–02	0.17611E–02	0.14950E–02
L_2	V1	0.76280E–02	0.75196E–02	0.58867E–02
	V2	0.53492E–02	0.53230E–02	0.43218E–02
	V3	0.49927E–02	0.50294E–02	0.40481E–02
L_∞	V1	0.90407E–01	0.84460E–01	0.72626E–01
	V2	0.62491E–01	0.62843E–01	0.55163E–01
	V3	0.56980E–01	0.56785E–01	0.47270E–01

Table 3
Errors of u for linear wave propagation problem in randomized grid with $A = 0.3$.

$A = 0.3$		S1	S2	S3
$\max(I_x^N , I_y^N , I_z^N)$		0.80606E–15	0.73219E–15	0.52614E–15
L_1	V1	0.45555E–02	0.44788E–02	0.33241E–02
	V2	0.28239E–02	0.28165E–02	0.20111E–02
	V3	0.25671E–02	0.26006E–02	0.17955E–02
L_2	V1	0.13548E–01	0.13383E–01	0.98583E–02
	V2	0.84581E–02	0.84547E–02	0.58798E–02
	V3	0.76473E–02	0.77480E–02	0.51448E–02
L_∞	V1	0.13837E+00	0.13101E+00	0.10977E+00
	V2	0.90781E–01	0.92510E–01	0.78112E–01
	V3	0.82397E–01	0.81654E–01	0.64272E–01

Table 4Errors of u for linear wave propagation problem in randomized grid with $A = 0.4$.

$A = 0.4$		S1	S2	S3
$\max(I_x^N , I_y^N , I_z^N)$		0.79046E–15	0.68669E–15	0.54120E–15
L_1	V1	0.71376E–02	0.70266E–02	0.50898E–02
	V2	0.42686E–02	0.42787E–02	0.27345E–02
	V3	0.38336E–02	0.39053E–02	0.23108E–02
L_2	V1	0.20931E–01	0.20746E–01	0.15145E–01
	V2	0.12783E–01	0.12821E–01	0.82311E–02
	V3	0.11487E–01	0.11670E–01	0.68770E–02
L_∞	V1	0.18432E+00	0.17713E+00	0.14644E+00
	V2	0.12137E+00	0.12470E+00	0.10402E+00
	V3	0.11161E+00	0.10886E+00	0.84649E–01

cancellation calculated with S3 are the smallest among the three forms of metrics on the randomized grids with a variety of perturbation amplitudes A . Although the forms of Jacobian have no effect on the SCL, i.e. the values of I_x^N , I_y^N and I_z^N only depend on the methods of calculating the metrics, different forms of Jacobian have quite different influences on the numerical errors of the flow fields.

The last column in Table 5 denotes the accuracy improvements from S1 and V1 to S3 and V3, especially the improvements are quite obvious for highly irregular grids. At the same time Table 5 also shows that the accuracy improvements become smaller as the grid getting smoother. This means that the SCMM for metrics and Jacobian may possess more advantages than the CMM on low-quality grids.

From Table 1 to Table 5, the results of the linear wave propagation problem show that S3 is better than both S1 and S2 in calculating the metrics for the randomized grids. At the same time V2 is better than V1, and V3 is the best on computing the Jacobian. The advantage of S3 and V3 can be also demonstrated from Fig. 8 which shows u contours with the same grid domain as illustrated in Fig. 7. Only the results computed in randomized grid with $A = 0.4$ are illustrated in Fig. 8 in order to condense the paper. The improvement of S3 and V3 over the other forms of metrics and Jacobian can be obviously observed from Fig. 8a to Fig. 8i, while the results calculated with S1 are almost equivalent to the results with S2 when the same form of Jacobian is used.

5.2. Isentropic moving vortex problems

The initial conditions for the isentropic moving vortex problems are

$$(u', v', w') = \varepsilon \tau e^{\alpha(1-\tau^2)} \left(\frac{y - y_c}{r}, -\frac{x - x_c}{r}, 0 \right),$$

$$T' = -\frac{(\gamma - 1)\varepsilon^2}{4\alpha\gamma} e^{2\alpha(1-\tau^2)}, \quad S' = 0,$$

$$\tau = \frac{r}{r_c}, \quad r = \sqrt{(x - x_c)^2 + (y - y_c)^2}.$$

The specific heat ratio of the fluid is set to $\gamma = 1.4$, and $r_c = 0.2$ denotes the vortex core width, $\varepsilon = 0.3$ represents vortex strength, $\alpha = 0.4$ is the parameter of the length scale of vortex decay. $T = p/\rho$ is the temperature implied that the free-stream Mach number is set to $M_\infty = 1/\sqrt{\gamma}$, and $S = p/\rho^\gamma$ is the entropy. The center of the vortex is initially located at $(x_c, y_c) = (0, 0)$. The mesh and the time marching method used in this test are the same as linear wave propagation ones.

Table 5Errors of u for linear wave propagation problem in randomized grids.

Randomized grids		$error_{V1S1}$	$error_{V3S3}$	$\frac{error_{V1S1}}{error_{V3S3}}$
L_1	$A = 0.1$	0.15167E–02	0.13588E–02	1.12
	$A = 0.2$	0.25907E–02	0.14950E–02	1.73
	$A = 0.3$	0.45555E–02	0.17955E–02	2.54
	$A = 0.4$	0.71376E–02	0.23108E–02	3.09
L_2	$A = 0.1$	0.41481E–02	0.35168E–02	1.18
	$A = 0.2$	0.76280E–02	0.40481E–02	1.88
	$A = 0.3$	0.13548E–01	0.51448E–02	2.63
	$A = 0.4$	0.20931E–01	0.68770E–02	3.04
L_∞	$A = 0.1$	0.48627E–01	0.34034E–01	1.43
	$A = 0.2$	0.90407E–01	0.47270E–01	1.91
	$A = 0.3$	0.13837E+00	0.64272E–01	2.15
	$A = 0.4$	0.18432E+00	0.84649E–01	2.18

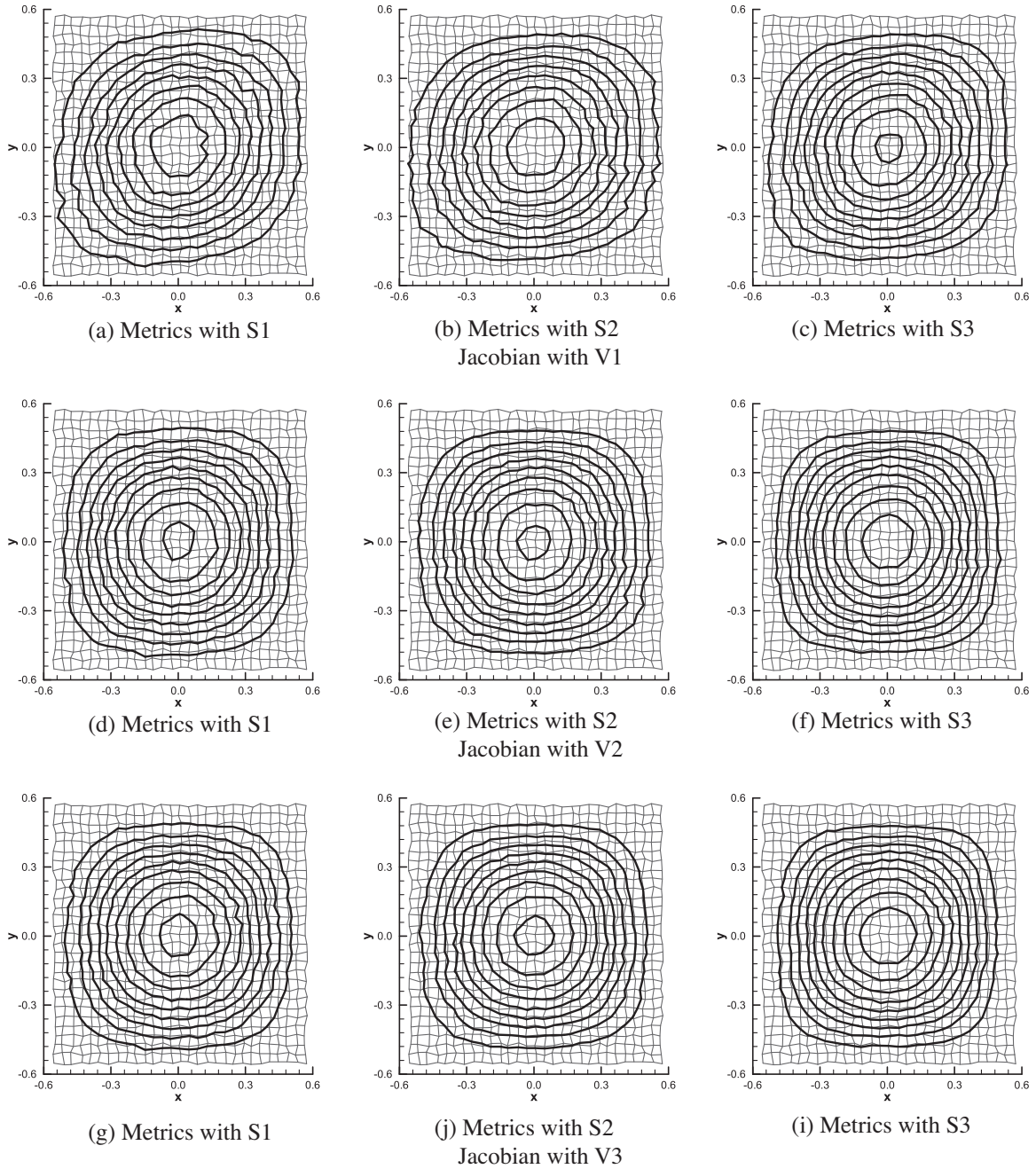


Fig. 8. Nine equally spaced contours from 0.1 (outer) to 0.9 (inner) of u calculated in randomized grid with $A = 0.4$.

The numerical results of the nonlinear Euler equation in randomized grids with sixth-order procedure are shown in Tables 6–9 with different perturbation amplitudes A . The equally spaced contours of ρ are showed in Fig. 9. The maximum of $|I_x^N|$, $|I_y^N|$ and $|I_z^N|$ in Tables 6–9 which are the same to those in Tables 1–4 are listed for completeness, and three error norms of ρ are included also. From Tables 6–9, the same conclusions can be gained for the nonlinear Euler equation as those for the linear wave propagation equation. The errors of ρ in Table 10 indicate that the combination of S3 and V3 is the best choice for metrics and Jacobian respectively. Similar conclusion can also be drawn from the previous test case as shown from Tables 1–5.

Fig. 9 shows the contours of density for different forms of metrics and Jacobian on the same randomized grid with $A = 0.4$. It is obvious that the combination of S3 and V3 can produce the best result. The error differences between V1S1 and V3S3 in Table 10 seem to be larger than those in Table 5 though the same randomized grids are used. In other words, on irregular

Table 6Errors of ρ for moving vortex problems in randomized grid with $A = 0.1$.

$A = 0.1$		S1	S2	S3
$\max \left(\left \rho_x^N \right , \left \rho_y^N \right , \left \rho_z^N \right \right)$		0.84184E–15	0.74457E–15	0.43117E–15
L_1	V1	0.16065E–03	0.13706E–03	0.11577E–03
	V2	0.11337E–03	0.97785E–04	0.76867E–04
	V3	0.96876E–04	0.10232E–03	0.69005E–04
L_2	V1	0.32358E–03	0.28872E–03	0.24060E–03
	V2	0.23129E–03	0.20632E–03	0.16137E–03
	V3	0.20108E–03	0.20698E–03	0.14111E–03
L_∞	V1	0.43806E–02	0.37414E–02	0.32572E–02
	V2	0.33676E–02	0.26946E–02	0.23176E–02
	V3	0.31954E–02	0.27164E–02	0.18761E–02

Table 7Errors of ρ for moving vortex problems in randomized grid with $A = 0.2$.

$A = 0.2$		S1	S2	S3
$\max \left(\left \rho_x^N \right , \left \rho_y^N \right , \left \rho_z^N \right \right)$		0.86827E–15	0.93648E–15	0.68993E–15
L_1	V1	0.34893E–03	0.31708E–03	0.25453E–03
	V2	0.24133E–03	0.21757E–03	0.16319E–03
	V3	0.20750E–03	0.21589E–03	0.14327E–03
L_2	V1	0.68171E–03	0.62903E–03	0.50559E–03
	V2	0.47771E–03	0.43526E–03	0.33007E–03
	V3	0.41792E–03	0.42640E–03	0.28700E–03
L_∞	V1	0.94541E–02	0.85188E–02	0.65796E–02
	V2	0.68400E–02	0.57185E–02	0.46375E–02
	V3	0.65495E–02	0.55125E–02	0.40477E–02

Table 8Errors of ρ for moving vortex problems in randomized grid with $A = 0.3$.

$A = 0.3$		S1	S2	S3
$\max \left(\left \rho_x^N \right , \left \rho_y^N \right , \left \rho_z^N \right \right)$		0.80606E–15	0.73219E–15	0.52614E–15
L_1	V1	0.56699E–03	0.54005E–03	0.42227E–03
	V2	0.38704E–03	0.36178E–03	0.26312E–03
	V3	0.33639E–03	0.34433E–03	0.22611E–03
L_2	V1	0.11004E–02	0.10500E–02	0.81511E–03
	V2	0.75421E–03	0.70358E–03	0.51495E–03
	V3	0.66489E–03	0.67050E–03	0.44323E–03
L_∞	V1	0.15476E–01	0.14236E–01	0.11194E–01
	V2	0.10233E–01	0.92216E–02	0.69364E–02
	V3	0.10048E–01	0.84305E–02	0.65072E–02

Table 9Errors of ρ for moving vortex problems in randomized grid with $A = 0.4$.

$A = 0.4$		S1	S2	S3
$\max \left(\left \rho_x^N \right , \left \rho_y^N \right , \left \rho_z^N \right \right)$		0.79046E–15	0.68669E–15	0.54120E–15
L_1	V1	0.81119E–03	0.79601E–03	0.61839E–03
	V2	0.54851E–03	0.52656E–03	0.37718E–03
	V3	0.48192E–03	0.48645E–03	0.31867E–03
L_2	V1	0.15903E–02	0.15568E–02	0.11793E–02
	V2	0.10689E–02	0.10186E–02	0.72158E–03
	V3	0.95062E–03	0.94747E–03	0.61404E–03
L_∞	V1	0.22677E–01	0.20641E–01	0.16874E–01
	V2	0.14402E–01	0.13207E–01	0.94600E–02
	V3	0.14379E–01	0.11777E–01	0.92103E–02

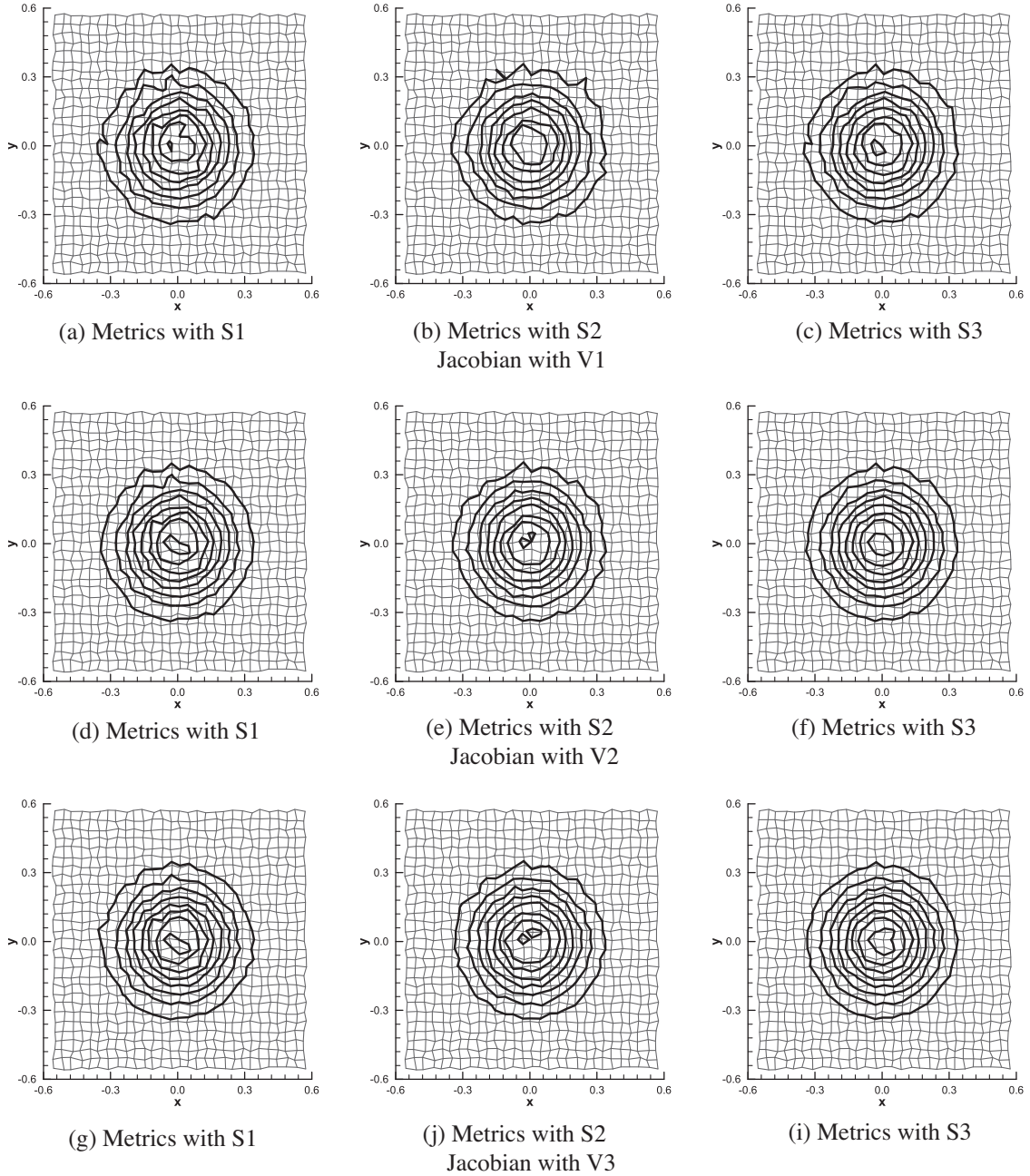


Fig. 9. Eight equally spaced contours of density from 0.92 (inner) to 0.99 (outer) calculated in randomized grid with $A = 0.4$.

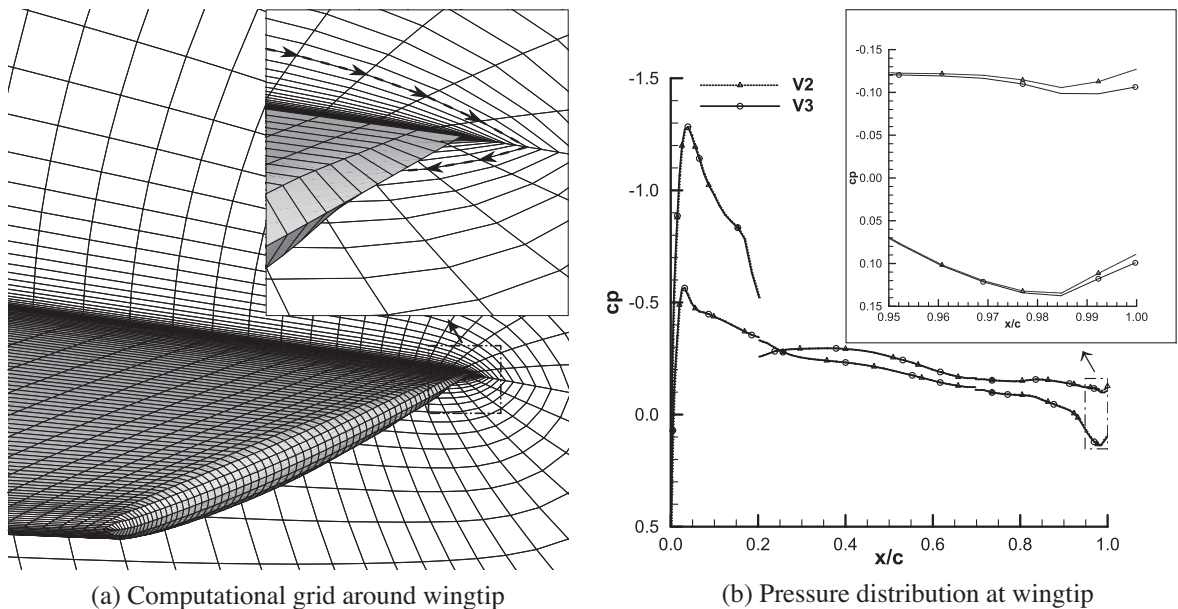
grids, the accuracy improvements benefited from S3 and V3 seems to be more evident for nonlinear problems than linear problems.

5.3. ONERA M6 wing

The current simulations is based on the flow conditions of Test 2308 of Ref. [20]. The free stream flow is characterized as $Ma_\infty = 0.8395$, $\alpha = 3.06^\circ$, $Re_c = 11.72 \times 10^6$. The Reynolds number Re_c is based on the mean aerodynamic chord $c = 0.64607m$. Fig. 10a shows the computational grid around wingtip of ONERA M6 wing as well as the magnified grid around rear edge of wingtip, and the total grid points are about 0.32 million. In this test case, the high-order WCNS-E-5 [15] scheme is applied to discretize the spatial derivatives, and the Spalart–Allmaras model [21] is adopted. Based on the

Table 10Errors of ρ for moving vortex problems in randomized grids.

Randomized grids		$error_{V1S1}$	$error_{V3S3}$	$\frac{error_{V1S1}}{error_{V3S3}}$
L_1	$A = 0.1$	0.16065E-03	0.69005E-04	2.33
	$A = 0.2$	0.34893E-03	0.14327E-03	2.44
	$A = 0.3$	0.56699E-03	0.22611E-03	2.51
	$A = 0.4$	0.81119E-03	0.31867E-03	2.55
L_2	$A = 0.1$	0.32358E-03	0.14111E-03	2.29
	$A = 0.2$	0.68171E-03	0.28700E-03	2.38
	$A = 0.3$	0.11004E-02	0.44323E-03	2.48
	$A = 0.4$	0.15903E-02	0.61404E-03	2.59
L_∞	$A = 0.1$	0.43806E-02	0.18761E-02	2.33
	$A = 0.2$	0.94541E-02	0.40477E-02	2.34
	$A = 0.3$	0.15476E-01	0.65072E-02	2.38
	$A = 0.4$	0.22677E-01	0.92103E-02	2.46

**Fig. 10.** Computational grid and pressure distribution of ONERA M6 wing.

SCMM derived in this paper, the symmetrical conservative form of metric S3 in Eq. (26) is utilized. To calculate the Jacobian, all three volume-like forms, i.e. V1, V2 and V3 derived in this paper are tested. As the traditional form of Jacobian V1 adopted, however, the calculation blows up, though the V1 Jacobians at every grid point are positive. The pressure distributions around wingtip calculated by Jacobian V2 and V3 are showed in Fig. 10b. We may see the difference of pressure distribution near the rear edge of wingtip, where the difference of V2 and V3 Jacobians $V^{23} = \frac{|V^2 - V^3|}{V^3}$ is obvious.

6. Concluding remarks

In FDM, there is a variety of equivalent differential forms of metrics and Jacobian which may introduce different discretization errors. In order to settle the problem on developing a proper form, a Symmetrical Conservative Metric Method (SCMM) is newly derived to calculate metrics and Jacobian. The SCMM is derived from the geometry analogy between FDM and FVM and is based on the CMM [1] which can ensure the SCL with the condition $\delta^1 = \delta^2$. The constraint $\delta^3 = \delta^2$, which is recommended by Deng et al. [1], is proved from geometry viewpoint.

In FDM, the metrics can be computed with various non-conservative or conservative forms, but only the symmetrical conservative form of metrics S3 can simultaneously produces the vectorized-surface-like metrics and ensures the SCL. The Jacobian can also be computed with various forms, but only the symmetrical conservative form of Jacobian V3 with the proper difference scheme can be enclosed by its vectorized surfaces. Then the SCMM is expressed as the combination of S3 and V3, together with $\delta_\xi^1 = \delta_\xi^2 = \delta_\xi^3$, $\delta_\eta^1 = \delta_\eta^2 = \delta_\eta^3$ and $\delta_\zeta^1 = \delta_\zeta^2 = \delta_\zeta^3$. The similarity between second-order FDM and FVM is essential for

that metrics and Jacobian calculated by the SCMM in FDM is identical to the vectorized surfaces and cell volume in FVM, respectively. The main advantage of the SCMM is that it makes the high-order FDM such as WCNS calculations around complex geometry flows possible and somewhat easy.

The numerical tests in the present work show that the SCMM can evidently increase the numerical accuracy and robustness on irregular grids with high-order WCNS schemes. Thus the SCMM is mightily suggested as a preferable method to calculate the metrics and Jacobian in high-order FDM simulations of flows around complex geometries.

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